

SIEM'S: HOME LAB

2110040122@klh.edu.in

BOBBA PANDURANGA

CYBER SECURITY

Table of Contents

Virtual Machine Installation	1
Installation of Windows 10 Pro in Virtual Machine	2
Type chapter title (level 3)	3
Malware detection	4
Wazuh for malware detection	5
Detecting malicious activities with threat detection	6

1. INSTALLING VIRTUAL MACHINE

STEP 1: Go To The Virtual Machine Official Website: <https://www.virtualbox.org/>



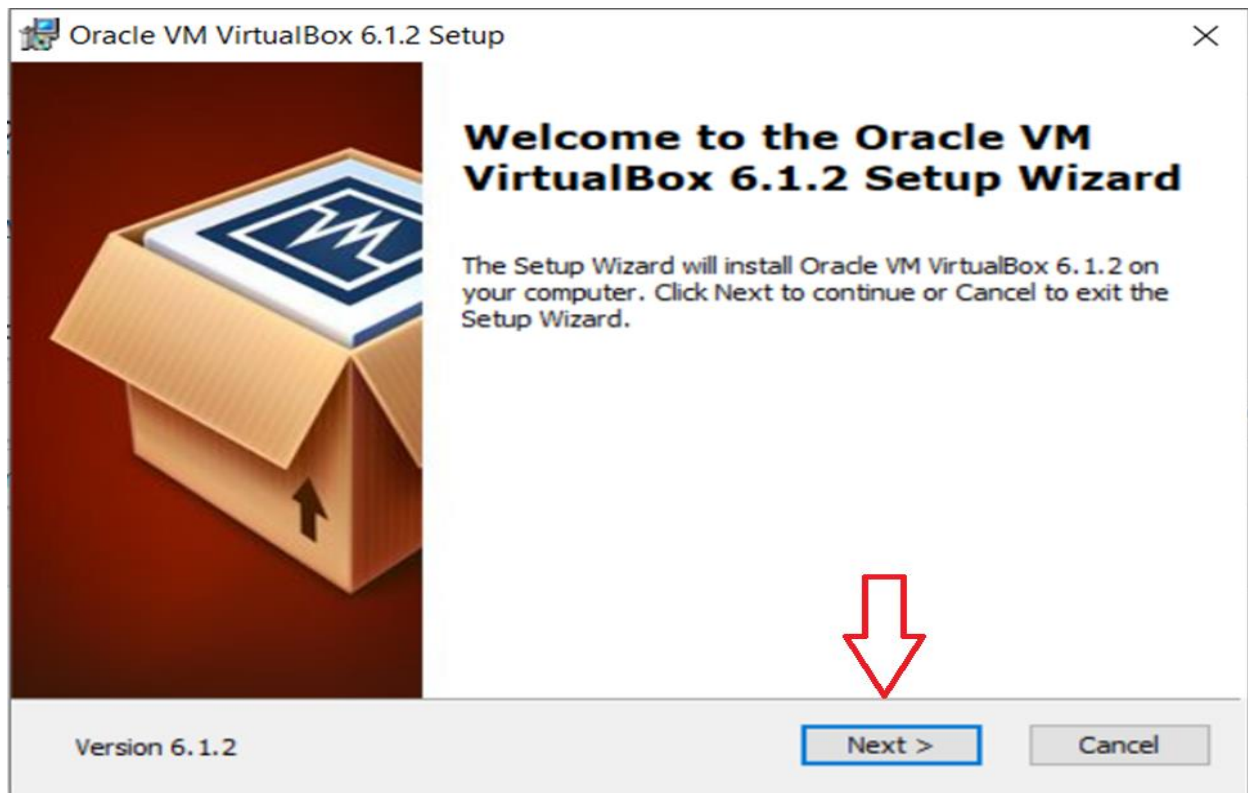
STEP 2: Download The Virtual Machine By Clicking On Download Button



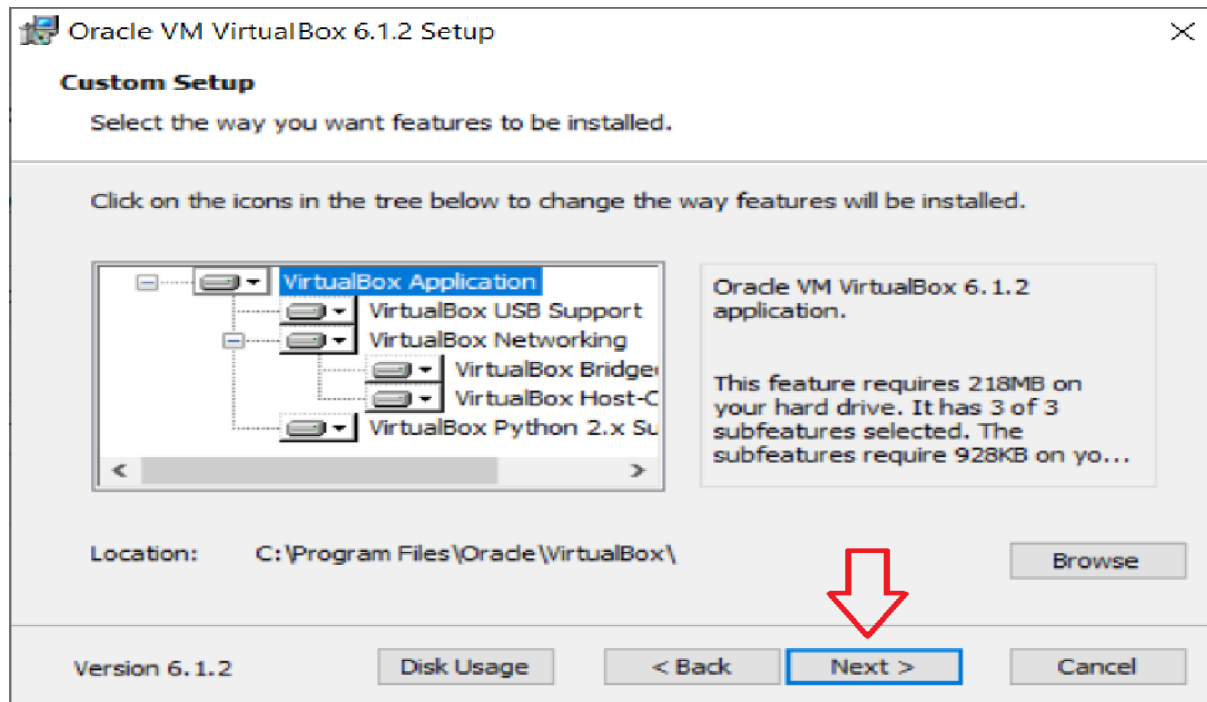
STEP 3: Select Virtualbox As Per Your OS



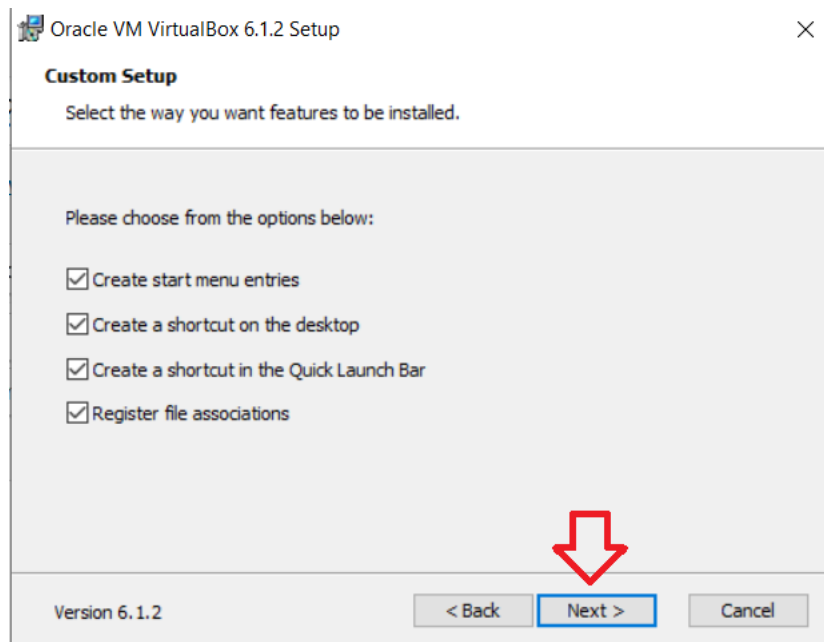
STEP 4: Install The Virtual Box



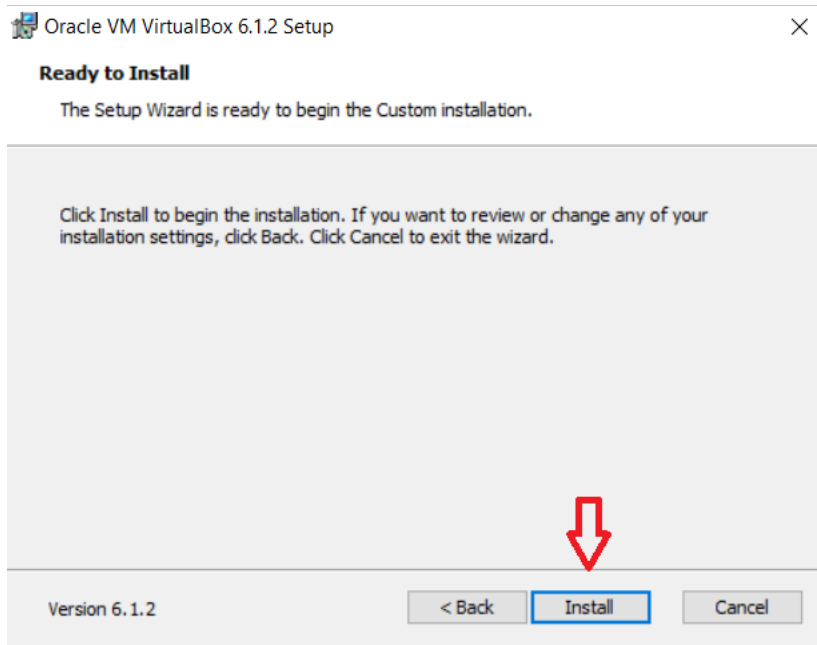
STEP 5: Select Installation Location



STEP 6: Creating Entries and Shortcuts:



STEP 7: Ready To Install:



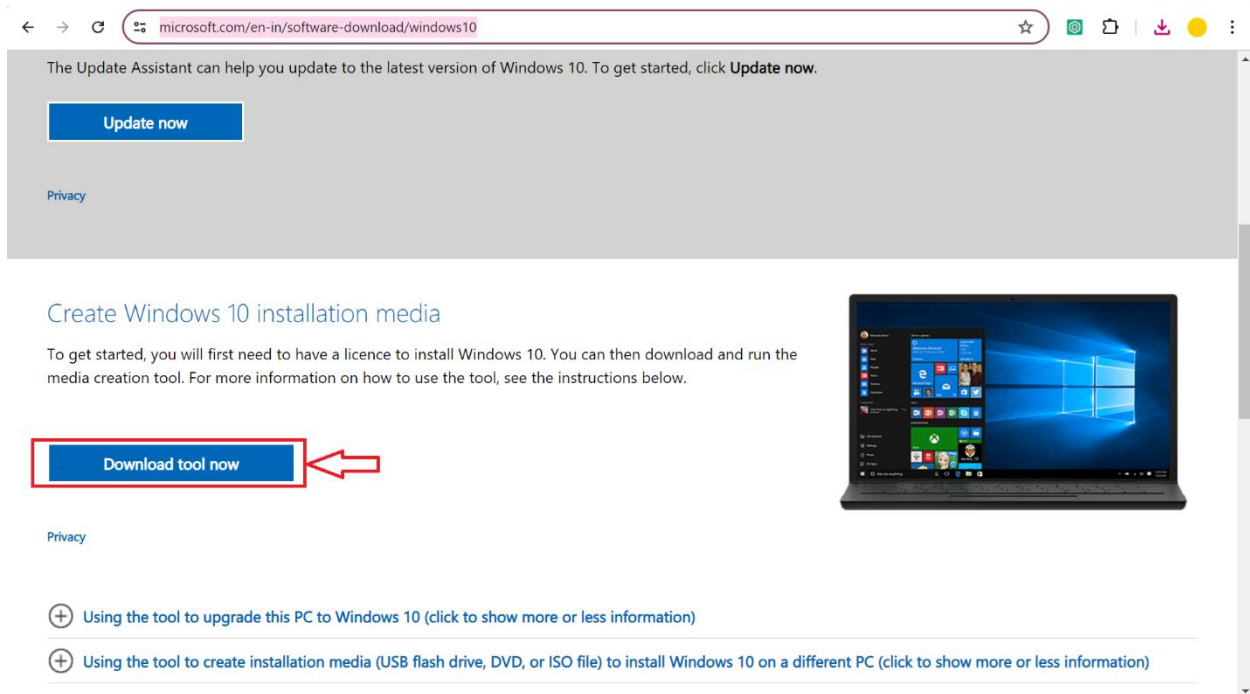
STEP 8: Finished Installation



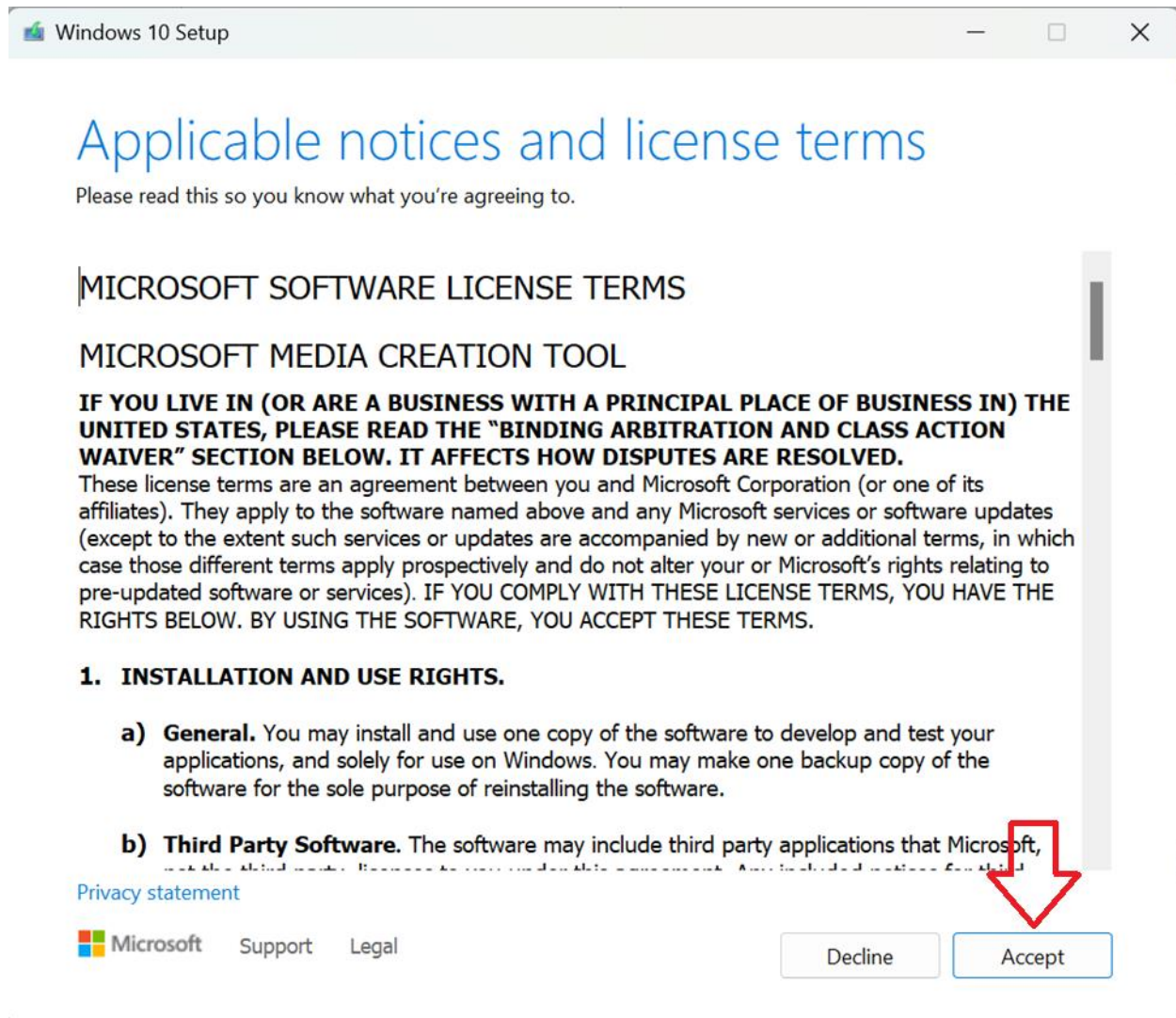
1.1 INSTALING WIN 10 PRO IN VIRTUAL MACHINE

STEP 1: GO TO WINDOWS OFFICIAL WEBSITE: <https://www.microsoft.com/en-in/software-download/windows10>

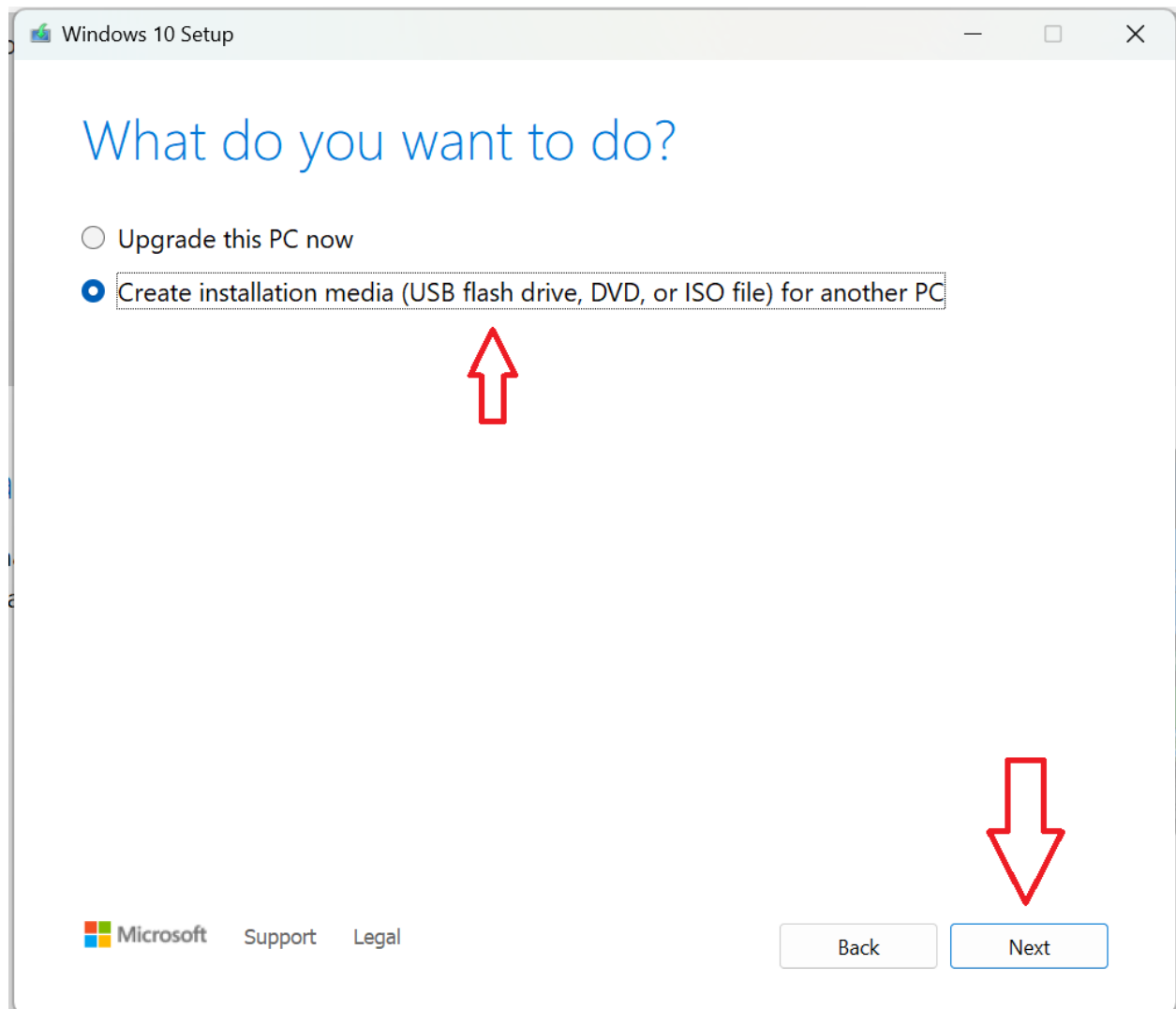
STEP 2: DOWNLOAD THE TOOL.



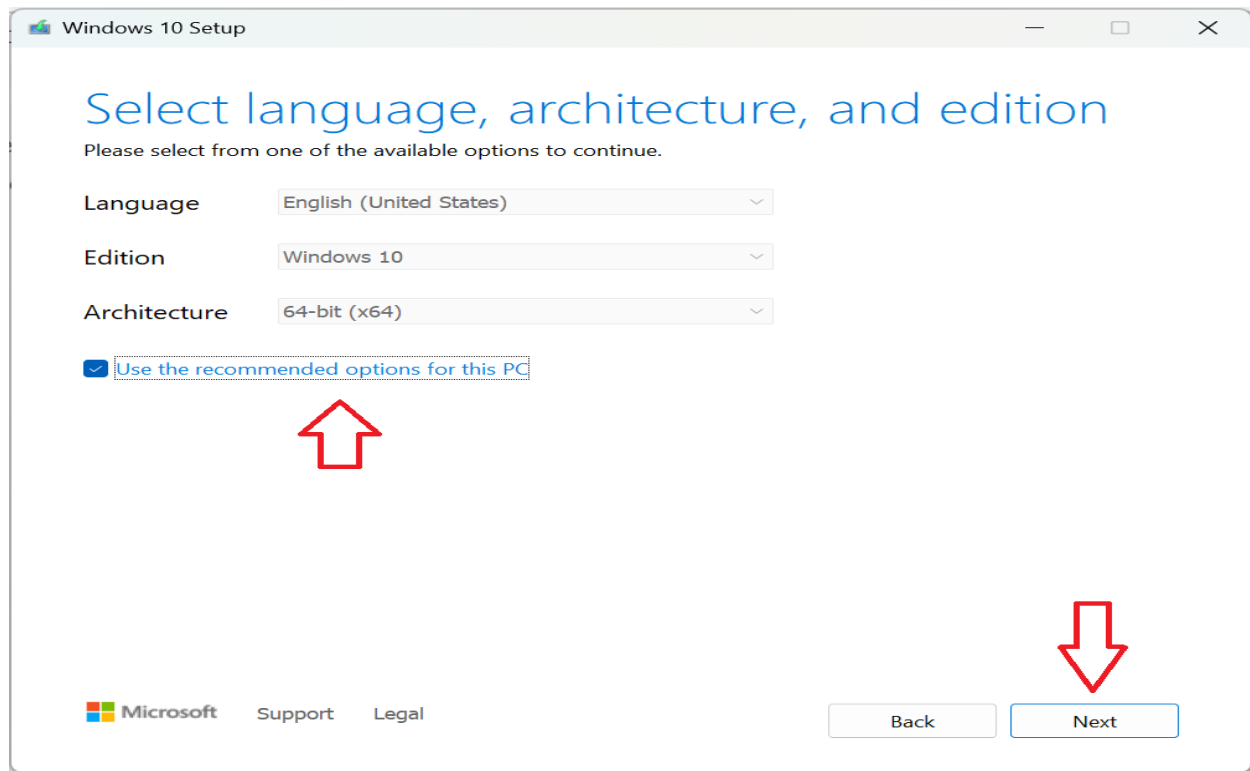
STEP 3: GO TO DOWNLOADS AND FIND THE TOOL AND RUN IT.



STEP 4: SELECT THE CREATE INSTALLATION MEDIA (ISO) AND THEN CLICK ON NEXT



STEP 5: SELECT THE OPTIONS AS SHOWN BELOW



The screenshot shows the 'Windows 10 Setup' window with the title bar 'Windows 10 Setup'. The main heading is 'Select language, architecture, and edition' in blue. Below it, a subtitle reads 'Please select from one of the available options to continue.' There are three dropdown menus: 'Language' set to 'English (United States)', 'Edition' set to 'Windows 10', and 'Architecture' set to '64-bit (x64)'. Below these is a checkbox labeled 'Use the recommended options for this PC' which is checked. A red arrow points up to this checkbox. At the bottom right, there are 'Back' and 'Next' buttons. A red arrow points down to the 'Next' button. At the bottom left, there is a Microsoft logo and links for 'Support' and 'Legal'.

Windows 10 Setup

Select language, architecture, and edition

Please select from one of the available options to continue.

Language: English (United States)

Edition: Windows 10

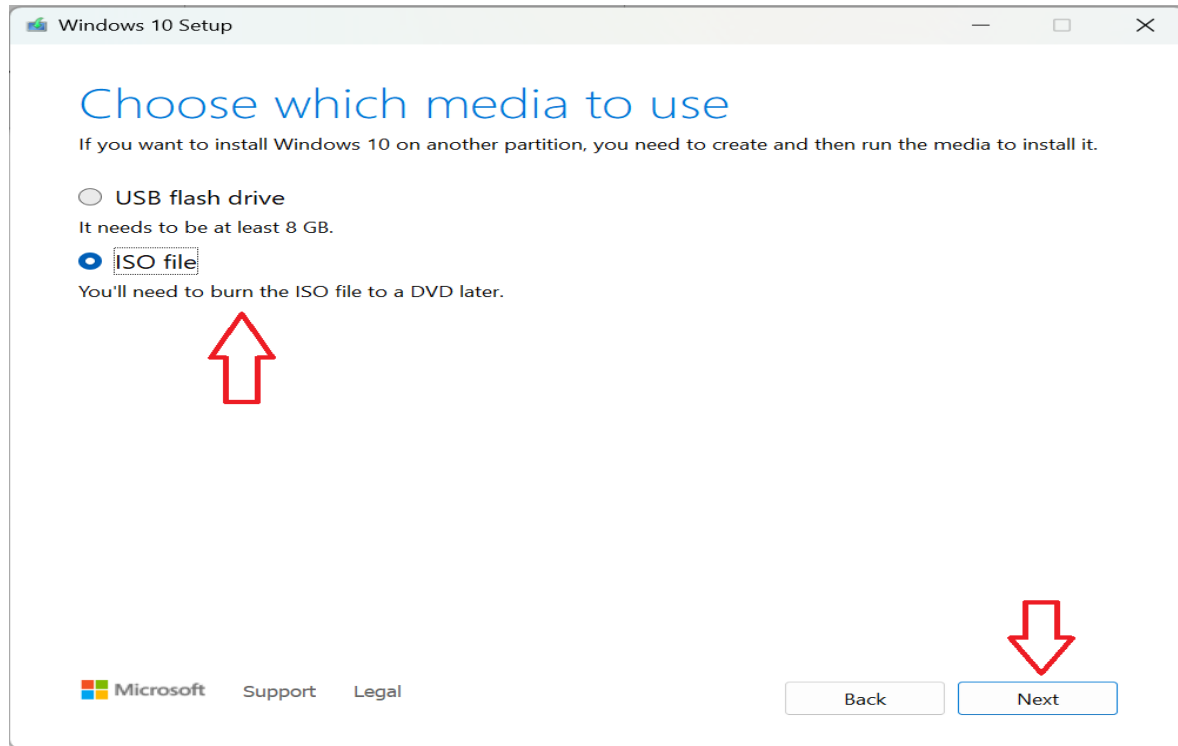
Architecture: 64-bit (x64)

☒ Use the recommended options for this PC

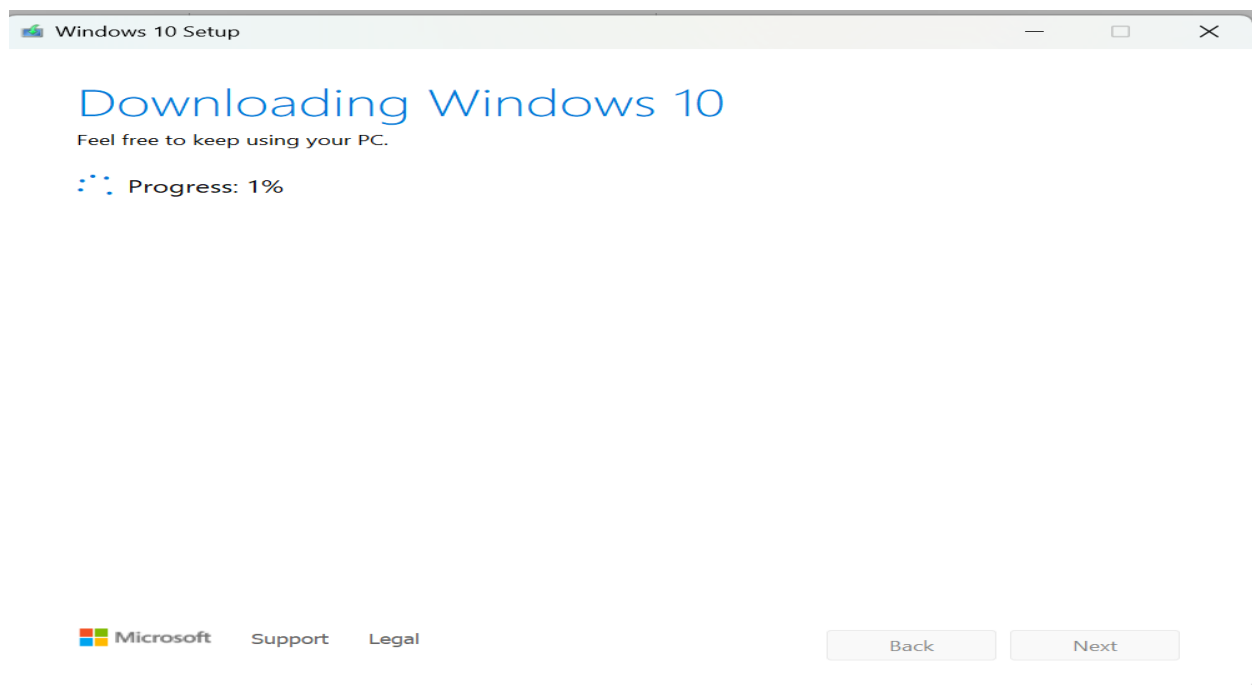
Microsoft Support Legal

Back Next

STEP 6: SELECT THE ISO FILE AND CLICK ON NEXT

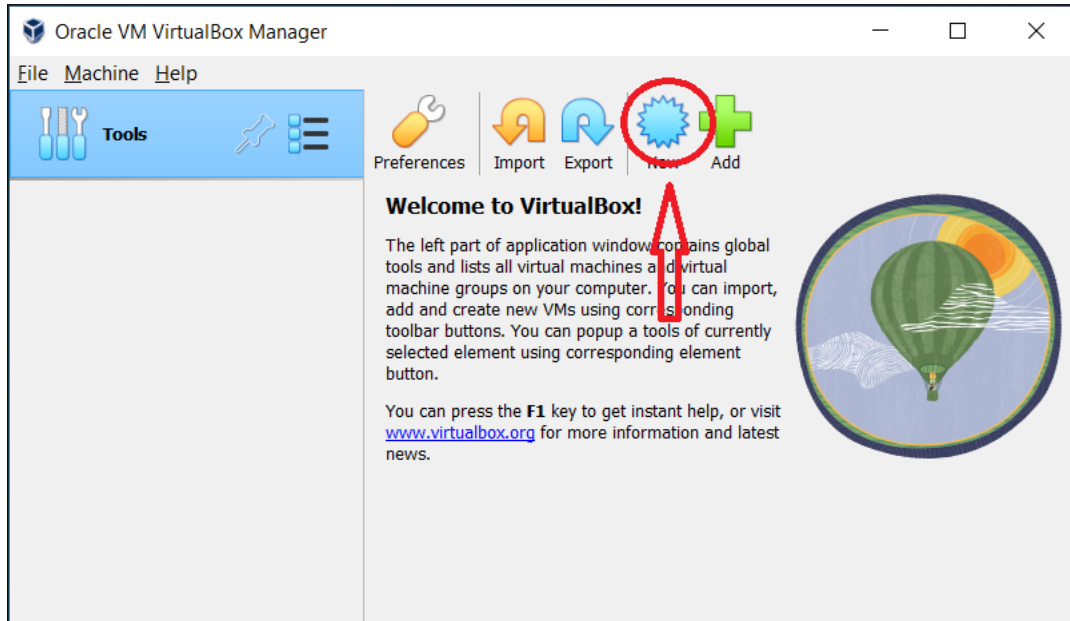


STEP 7: SELECT THE FILE DESTINATION AND CLICK ON NEXT WIN 10 FILE WILL START DOWNLOADING



INSTALING THE WIN 10 IN VIRTUALBOX

STEP 1: OPEN THE VIRTUALBOX AND SELECT THE NEW OPTION

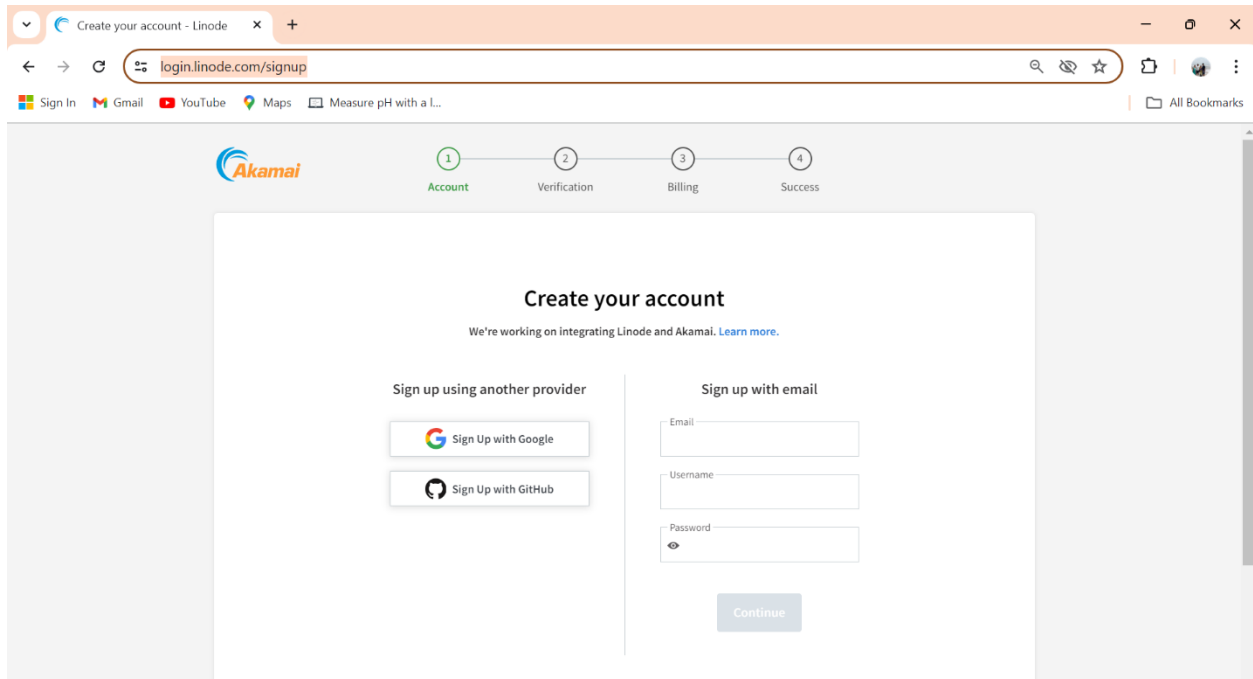


STEP 2: SELECT THE WIN 10 ISO FILE AND CLICK ON FINISH

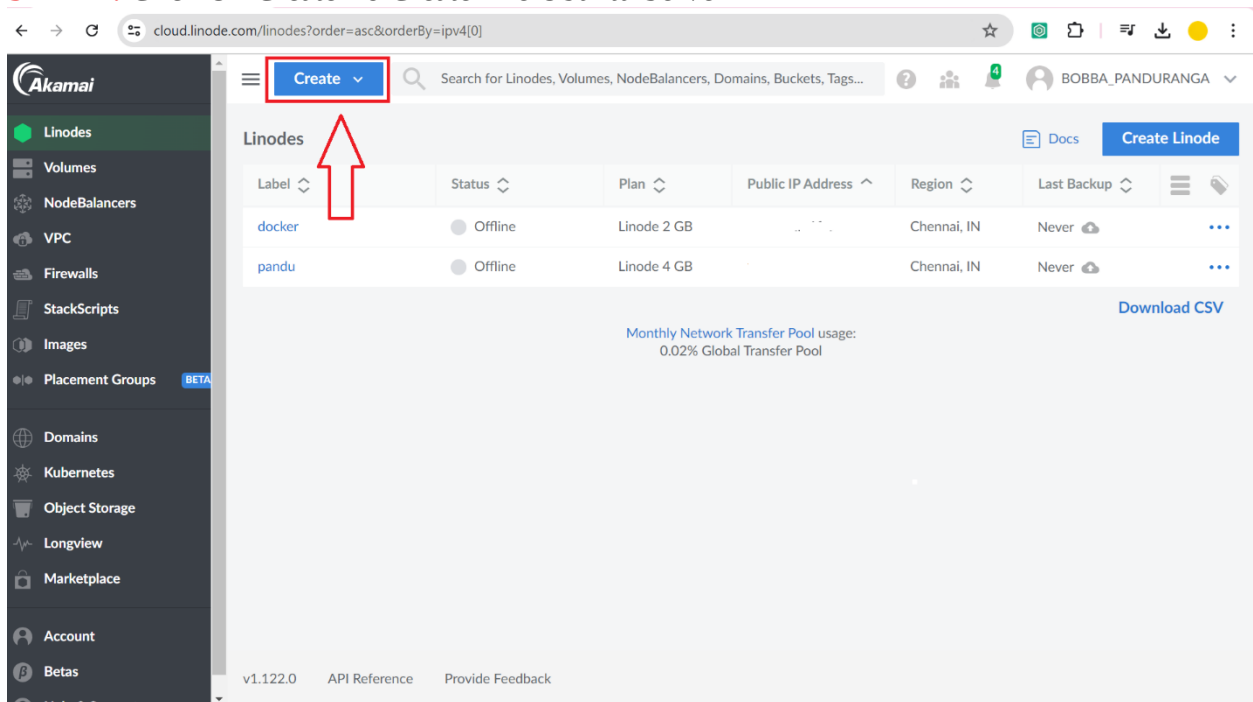
STEP 3: That is Windows 10 has been installed on the Virtual Machine. You can now open the Windows 10 operating system in VM. Set up a new username and password and use Windows 10 in VM for various purposes

STAGE 3 : CREATING ubuntu SERVER AND DOCKER ON akamai cloud manger

STEP 1: Go to This Website <https://Login.Linode.Com/Signup> And Signup First And Create Your Account



STEP 2: Click On Create To Create The Ubuntu Server

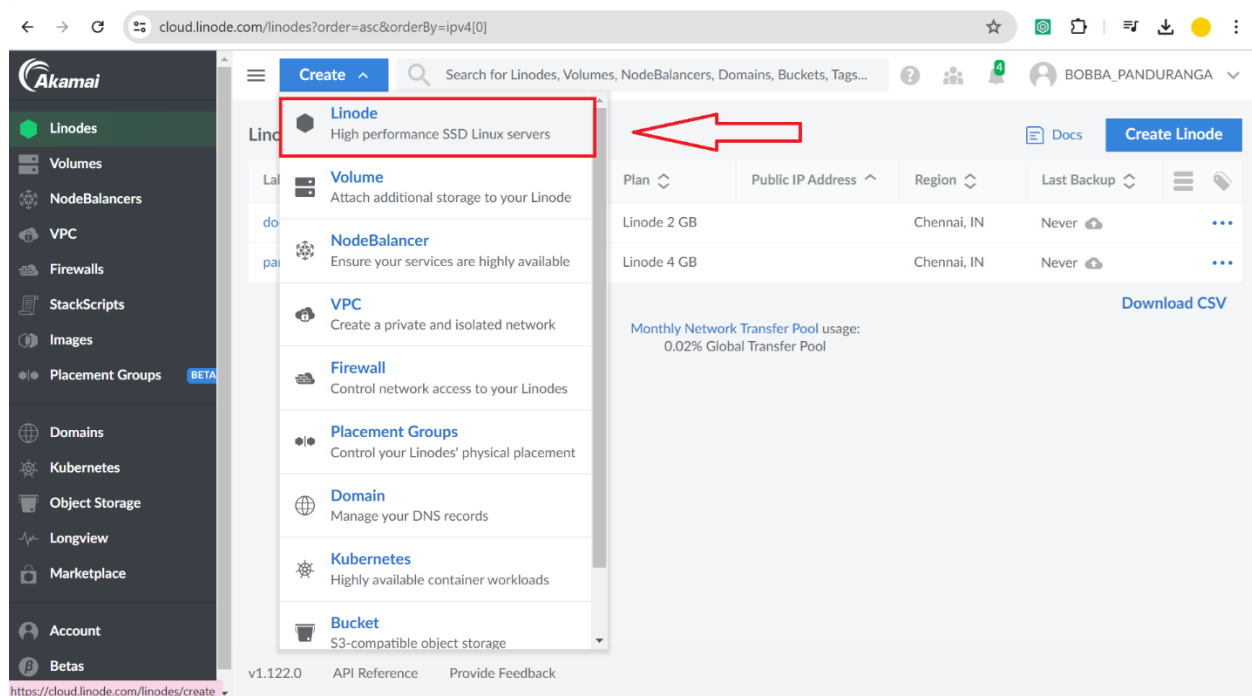


Label	Status	Plan	Public IP Address	Region	Last Backup
docker	Offline	Linode 2 GB		Chennai, IN	Never
pandu	Offline	Linode 4 GB		Chennai, IN	Never

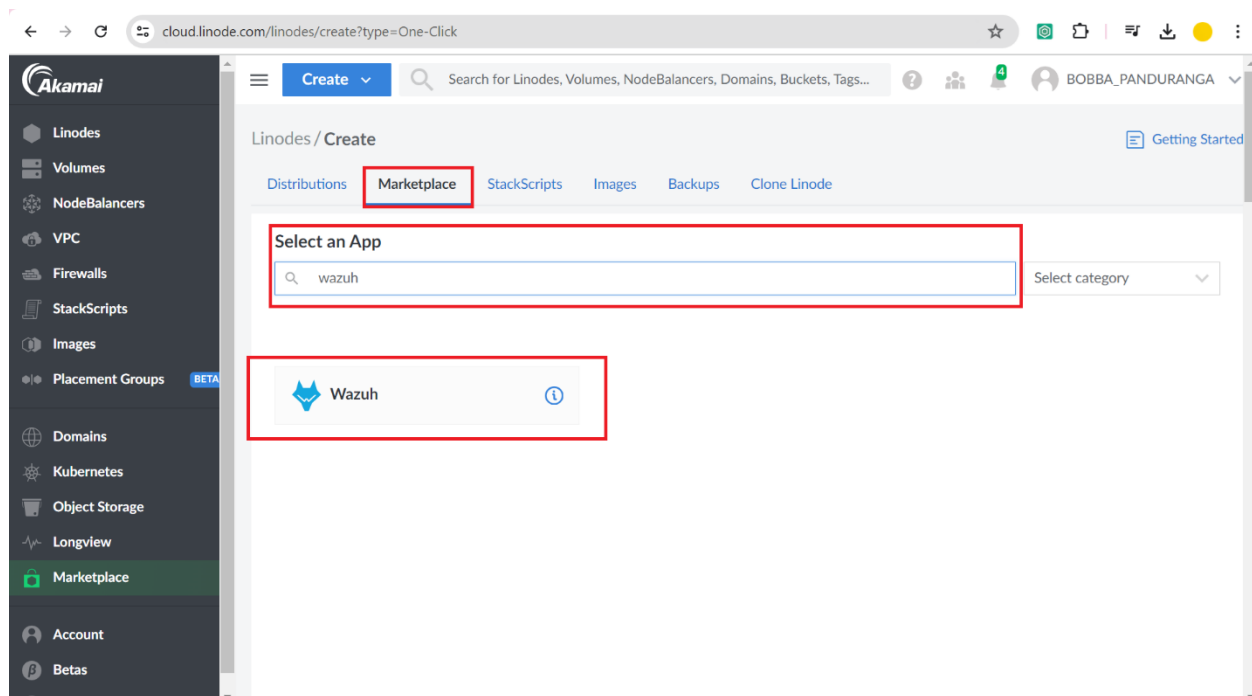
Monthly Network Transfer Pool usage:
0.02% Global Transfer Pool

Download CSV

STEP 3: Click On Linode



STEP 4: Select The Marketplace → Search Wazuh On Select An App → Select Wazuh:



STEP 5: Scroll Down And Fill The Details As Shown Below

cloud.linode.com/linodes/create?type=One-Click

Wazuh Setup

Email address (for the Let's Encrypt SSL certificate) (required)
xx
user@domain.tld

The limited sudo user to be created for the Linode: "No Capital ..."
xx

Advanced Options

These fields are additional configuration options and are not required for creation.

Disable root access over SSH?

☐ Yes

☒ No

Your Linode API token. This is needed to create your Wazuh server's DNS records

?

Weak

Subdomain

The subdomain for the DNS record: www (Requires Domain)

STEP 6: Select The Image File Make Sure That You Should Select The Ubuntu 22.04 LTS Only And Select Your Nearest Region

cloud.linode.com/linodes/create?type=One-Click

Select an Image

Images

X ▼

Region

You can use [our speedtest page](#) to find the best region for your current location.

Region

▼

Linode Plan

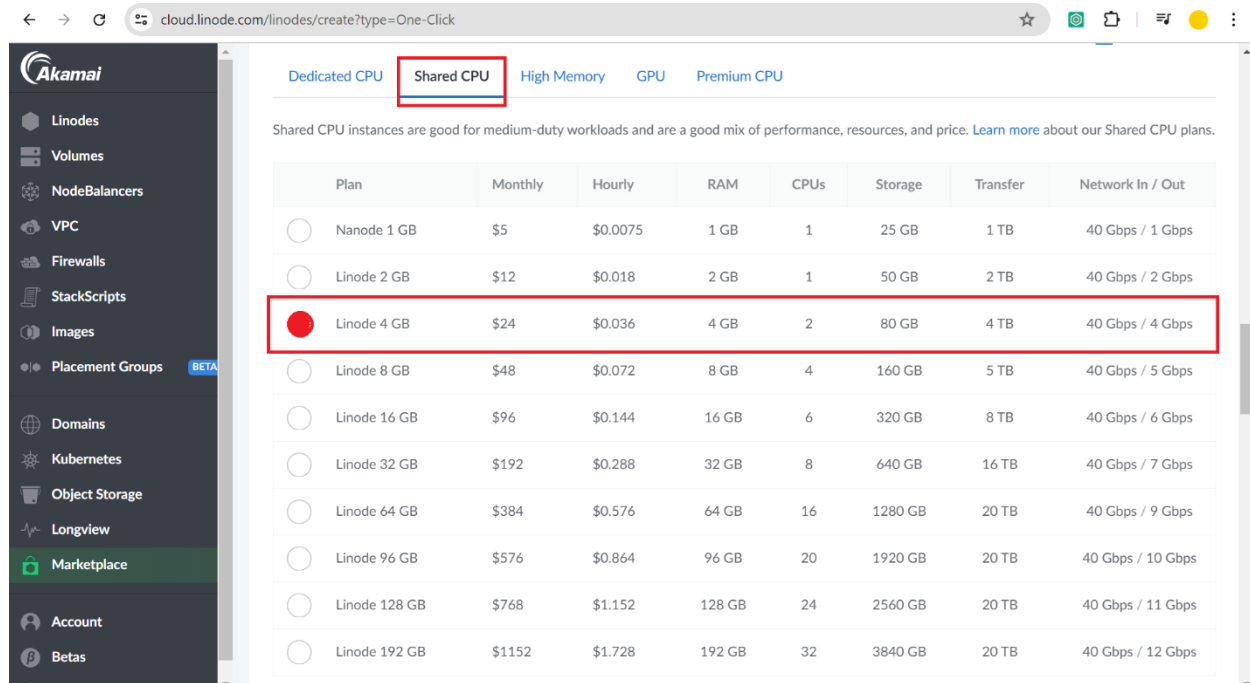
[Choosing a Plan](#)

Dedicated CPU [Shared CPU](#) [High Memory](#) [GPU](#) [Premium CPU](#)

Dedicated CPU instances are good for full-duty workloads where consistent performance is important. [Learn more](#) about our Dedicated CPU plans.

Plan	Monthly	Hourly	RAM	CPUs	Storage	Transfer	Network In / Out
Select a region to view plans and prices.							

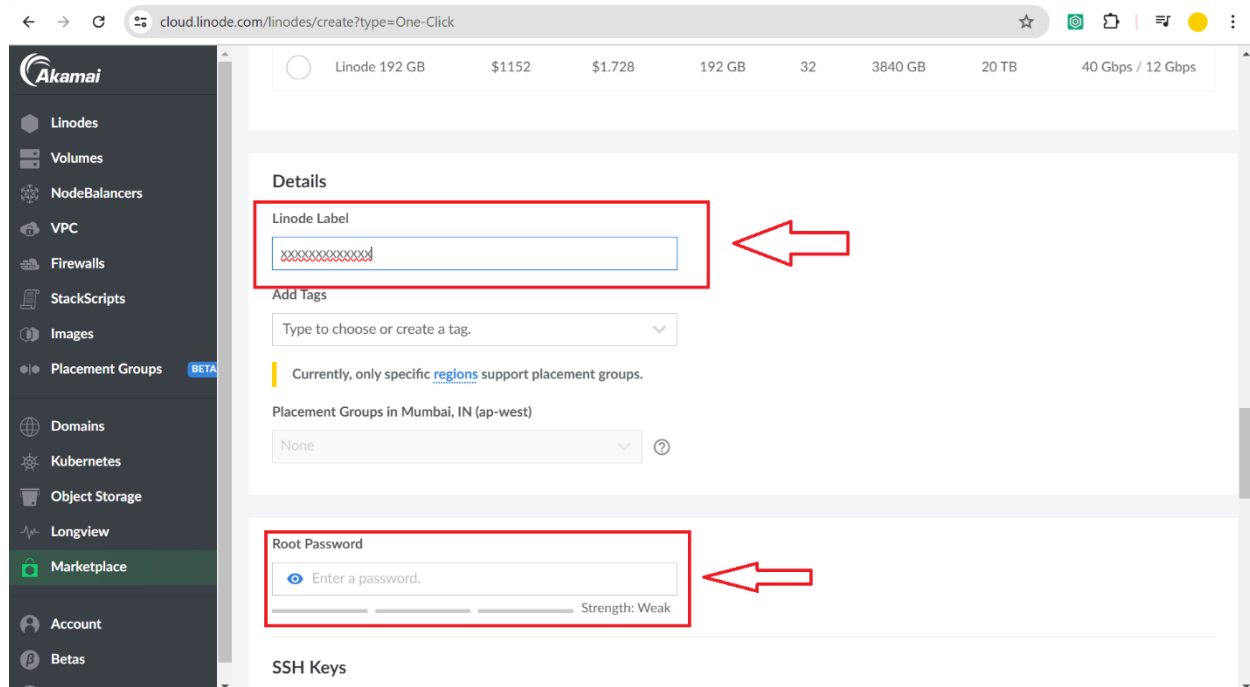
STEP 7::Select The Shared Cpu And Select The Linode 4 GB Plane As Shown Below



The screenshot shows the Linode 'Create Linode' page. The 'Shared CPU' tab is selected and highlighted with a red box. Below the tabs, a table lists various Linode plans. The 'Linode 4 GB' plan is highlighted with a red box.

	Plan	Monthly	Hourly	RAM	CPUs	Storage	Transfer	Network In / Out
☐	Nanode 1 GB	\$5	\$0.0075	1 GB	1	25 GB	1 TB	40 Gbps / 1 Gbps
☐	Linode 2 GB	\$12	\$0.018	2 GB	1	50 GB	2 TB	40 Gbps / 2 Gbps
☒	Linode 4 GB	\$24	\$0.036	4 GB	2	80 GB	4 TB	40 Gbps / 4 Gbps
☐	Linode 8 GB	\$48	\$0.072	8 GB	4	160 GB	5 TB	40 Gbps / 5 Gbps
☐	Linode 16 GB	\$96	\$0.144	16 GB	6	320 GB	8 TB	40 Gbps / 6 Gbps
☐	Linode 32 GB	\$192	\$0.288	32 GB	8	640 GB	16 TB	40 Gbps / 7 Gbps
☐	Linode 64 GB	\$384	\$0.576	64 GB	16	1280 GB	20 TB	40 Gbps / 9 Gbps
☐	Linode 96 GB	\$576	\$0.864	96 GB	20	1920 GB	20 TB	40 Gbps / 10 Gbps
☐	Linode 128 GB	\$768	\$1.152	128 GB	24	2560 GB	20 TB	40 Gbps / 11 Gbps
☐	Linode 192 GB	\$1152	\$1.728	192 GB	32	3840 GB	20 TB	40 Gbps / 12 Gbps

STEP 8:Now Fill The Details Linode Label As You Like And Give The Root Password As You Like Make Sure That You Remember That Root Password Again You Cant Able To Change



The screenshot shows the 'Details' section of the Linode 'Create Linode' page. The 'Linode Label' field is highlighted with a red box and an arrow. The 'Root Password' field is also highlighted with a red box and an arrow. The 'Root Password' field shows a strength indicator of 'Weak'.

Details

Linode Label
xxxxxxxxxxxxxx

Add Tags
Type to choose or create a tag.

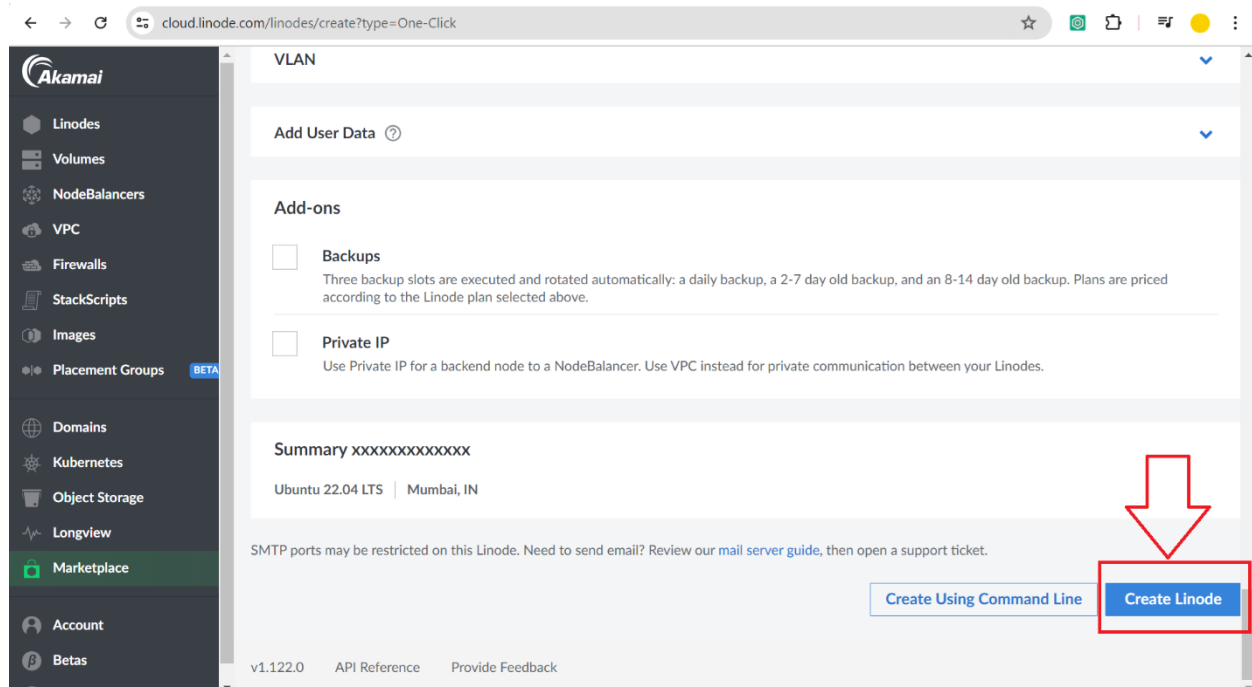
Currently, only specific [regions](#) support placement groups.

Placement Groups in Mumbai, IN (ap-west)
None

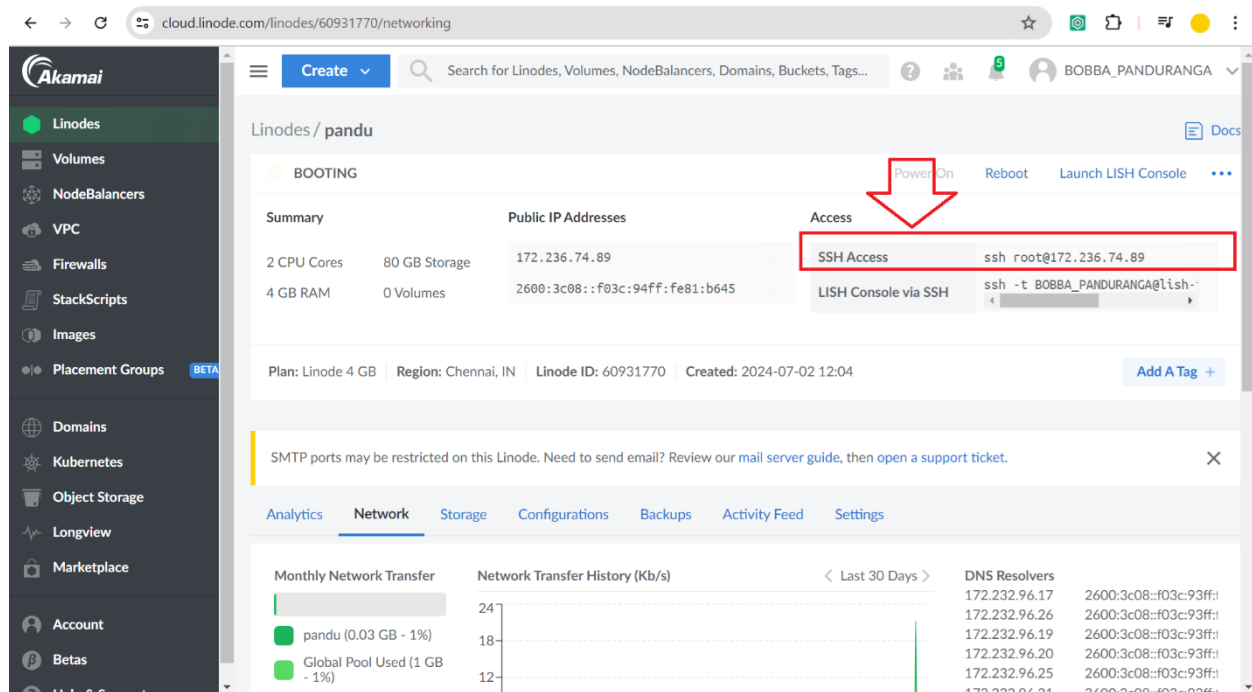
Root Password
Enter a password. Strength: Weak

SSH Keys

STEP 9: Click On Create The Linode Now You Are Good To Go



STEP 10: After creating the server now copy the SSH ACCESS CODE



STEP 11: GO TO CMD IN WINDOWS PAST THE SSH ACCESS CODE And Enter The Password As You Given Before

```
root@172-236-74-89: ~  
Microsoft Windows [Version 10.0.22631.3810]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\bobba>ssh root@172.236.74.89  
root@172.236.74.89's password:  
Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 5.15.0-113-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Wed Jul  3 02:37:38 PM UTC 2024  
  
System load:        0.16  
Usage of /:          9.0% of 78.17GB  
Memory usage:       75%  
Swap usage:          0%  
Processes:          131  
Users logged in:    0  
IPv4 address for eth0: 172.236.74.89  
IPv6 address for eth0: 2600:3c08::f03c:94ff:fe81:b645  
  
Expanded Security Maintenance for Applications is not enabled.  
  
3 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

STE:P12 to see the internal process such as

install

```
root@172-236-74-89: ~  
Usage of /:          9.0% of 78.17GB  
Memory usage:       75%  
Swap usage:          0%  
Processes:          131  
Users logged in:    0  
IPv4 address for eth0: 172.236.74.89  
IPv6 address for eth0: 2600:3c08::f03c:94ff:fe81:b645  
  
Expanded Security Maintenance for Applications is not enabled.  
  
3 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
*****  
Akamai Connected Cloud Wazuh Marketplace App  
App URL: https://172-236-74-89.ip.linodeusercontent.com  
Credentials File: /home/pandu/.credentials  
Documentation: https://www.linode.com/docs/products/tools/marketplace/guides/wazuh/  
*****  
To delete this message of the day: rm /etc/motd  
Last login: Tue Jul  2 12:06:16 2024 from 136.185.203.70  
root@172-236-74-89:~#  
root@172-236-74-89:~#  
root@172-236-74-89:~#  
root@172-236-74-89:~# htop
```

```

root@172-236-74-89: ~
0[|] 2.6%] Tasks: 40, 202 thr; 1 running
1[ 0.0%] Load average: 0.01 0.20 0.15
Mem[|] 2.80G/3.82G] Uptime: 00:06:50
Swp[|] 1.25M/512M]

  PID USER      PRI  NI  VIRT   RES   SHR  S  CPU% MEM%   TIME+  Command
 1116 root        20    0  112M  5828  4544  S   0.7  0.1   0:01.43 /var/ossec/bin/wazuh-authd
 1118 root        20    0  112M  5828  4544  S   0.7  0.1   0:01.41 /var/ossec/bin/wazuh-authd
 2717 root        20    0   8688  4732  3432  R   0.7  0.1   0:00.05 htop
    1 root        20    0   98M  11280  8036  S   0.0  0.3   0:01.13 /sbin/init
 392 root        19   -1 40036 18668 17588  S   0.0  0.5   0:00.17 /lib/systemd/systemd-journald
 439 root        20    0 25880  6636  4540  S   0.0  0.2   0:00.12 /lib/systemd/systemd-udev
 461 systemd-t  20    0 89364  6436  5632  S   0.0  0.2   0:00.05 /lib/systemd/systemd-timesyncd
 471 root        20    0   8732  4964  1768  S   0.0  0.1   0:00.19 /usr/sbin/haveged --Foreground --verbose=1
 501 systemd-t  20    0 89364  6436  5632  S   0.0  0.2   0:00.00 /lib/systemd/systemd-timesyncd
 565 systemd-n  20    0 16128  7920  6900  S   0.0  0.2   0:00.03 /lib/systemd/systemd-networkd
 583 systemd-r  20    0 25540 12592  8396  S   0.0  0.3   0:00.09 /lib/systemd/systemd-resolved
 676 root        20    0   6896  2836  2588  S   0.0  0.1   0:00.00 /usr/sbin/cron -f -P
 677 messagebu  20    0   8616  4476  3804  S   0.0  0.1   0:00.02 @dbus-daemon --system --address=systemd: --nofork --nop
 682 root        20    0 1253M 43536 35548  S   0.0  1.1   0:00.15 /usr/share/filebeat/bin/filebeat --environment systemd
 685 root        20    0 32612 19004 10376  S   0.0  0.5   0:00.07 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-sta
 687 syslog     20    0   217M  5016  3928  S   0.0  0.1   0:00.02 /usr/sbin/rsyslogd -n -iNONE
 693 root        20    0 1253M 43536 35548  S   0.0  1.1   0:00.01 /usr/share/filebeat/bin/filebeat --environment systemd
 694 root        20    0 1253M 43536 35548  S   0.0  1.1   0:00.03 /usr/share/filebeat/bin/filebeat --environment systemd
 695 root        20    0 1253M 43536 35548  S   0.0  1.1   0:00.00 /usr/share/filebeat/bin/filebeat --environment systemd
 696 root        20    0 1253M 43536 35548  S   0.0  1.1   0:00.00 /usr/share/filebeat/bin/filebeat --environment systemd
 707 root        20    0 15336  7420  6472  S   0.0  0.2   0:00.05 /lib/systemd/systemd-logind
 709 wazuh-das    20    0 1020M 186M 36444  S   0.0  4.8   0:07.89 /usr/share/wazuh-dashboard/node/bin/node --no-warnings

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice -F8Nice +F9Kill F10Quit

```

- ❑ Use the `htpasswd` command to reset the password for the Wazuh web interface. Replace `admin` with the appropriate username if different.

bash

Copy code

```
htpasswd -c /var/ossec/api/configuration/auth/user admin
```

- You will be prompted to enter and confirm the new password.

❑ Restart Wazuh API:

- After resetting the password, restart the Wazuh API service to apply the changes.

bash

Copy code

```
systemctl restart wazuh-api
```

❑

STEP 13: If Want To Change The Passwaord Of Wazuh Login Tou Can Do By Apply Ing The Following Comand

```
root@172-236-74-89: ~  
Memory usage:      75%  
Swap usage:        0%  
Processes:         131  
Users logged in:   0  
IPv4 address for eth0: 172.236.74.89  
IPv6 address for eth0: 2600:3c08::f03c:94ff:fe81:b645  
  
Expanded Security Maintenance for Applications is not enabled.  
  
3 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
*****  
Akamai Connected Cloud Wazuh Marketplace App  
App URL: https://172-236-74-89.ip.linodeusercontent.com  
Credentials File: /home/pandu/.credentials  
Documentation: https://www.linode.com/docs/products/tools/marketplace/guides/wazuh/  
*****  
To delete this message of the day: rm /etc/motd  
Last login: Tue Jul 2 12:06:16 2024 from 136.185.203.70  
root@172-236-74-89:~#  
root@172-236-74-89:~#  
root@172-236-74-89:~#  
root@172-236-74-89:~#  
root@172-236-74-89:~# htop  
root@172-236-74-89:~# htpasswd -c /var/ossec/api/configuration/auth/user admin
```

STEP 14: Copy The Ip As I Shown Below And Past It In New Tab Url Box

The screenshot shows the Akamai Linodes dashboard. On the left is a sidebar with navigation links: Linodes, Volumes, NodeBalancers, VPC, Firewalls, StackScripts, Images, Placement Groups, Domains, Kubernetes, Object Storage, Longview, Marketplace, Account, Betas, and Help & Support. The main content area displays a graph of network usage, a table of firewalls (showing none are assigned), and a table of IP addresses. The IP address table has columns for Address, Type, Default Gateway, Subnet Mask, Reverse DNS, and actions (Delete, Edit RDNS). A red box highlights the Reverse DNS field for the IP 172.236.74-89.ip.linodeusercontent.com. A red arrow points to this box. Below the table, there are links for v1.122.0, API Reference, and Provide Feedback.

Address	Type	Default Gateway	Subnet Mask	Reverse DNS	Actions
172.236.74.89	IPv4 - Public	172.236.74.1	255.255.255.0	172-236-74-89.ip.linodeusercontent.com	Delete Edit RDNS
fe80::f03c:94ff:fe81:b645	IPv6 - Link Local	fe80::1	ffff:ffff:ffff::		
2600:3c08::f03c:94ff:fe81:b645	IPv6 - SLAAC	fe80::1	ffff:ffff:ffff::		Edit RDNS

wazuh.

The Open Source Security Platform

 Username

 Password

Enter the password

Log In

Enter the admin



W.

Overview

a



wazuh.

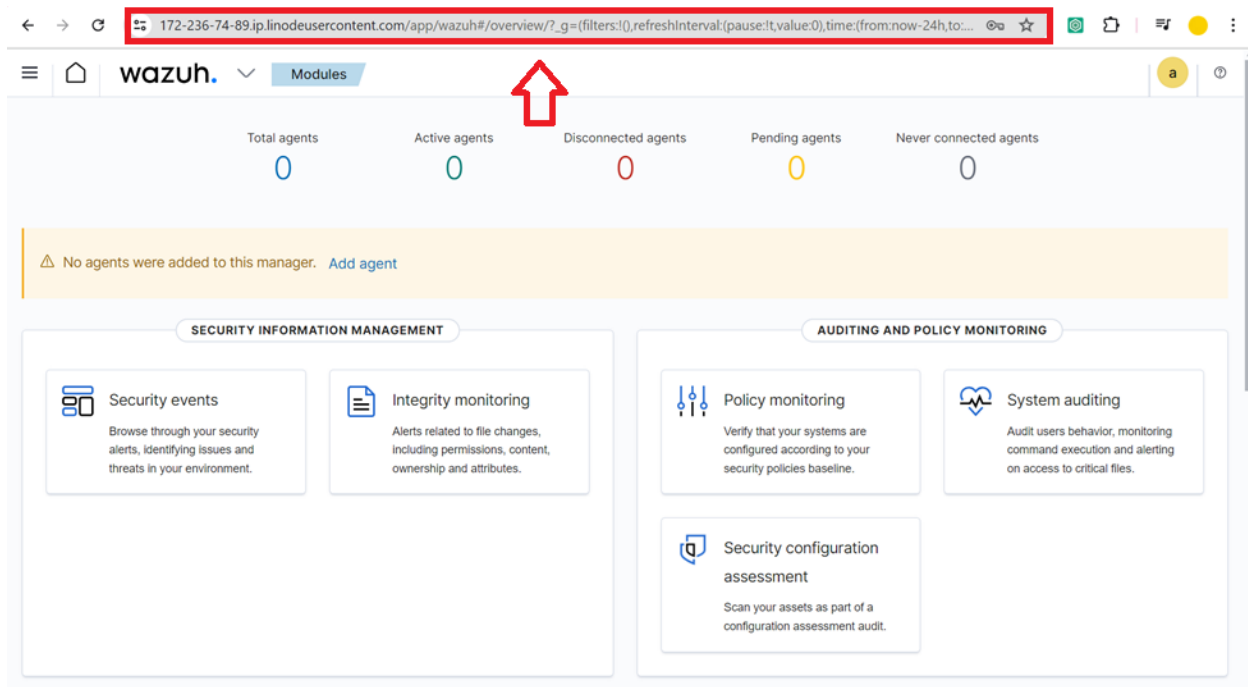
Check Wazuh API connection ✓

Check Wazuh API version ○

Check alerts index pattern ✓

Check monitoring index pattern ✓

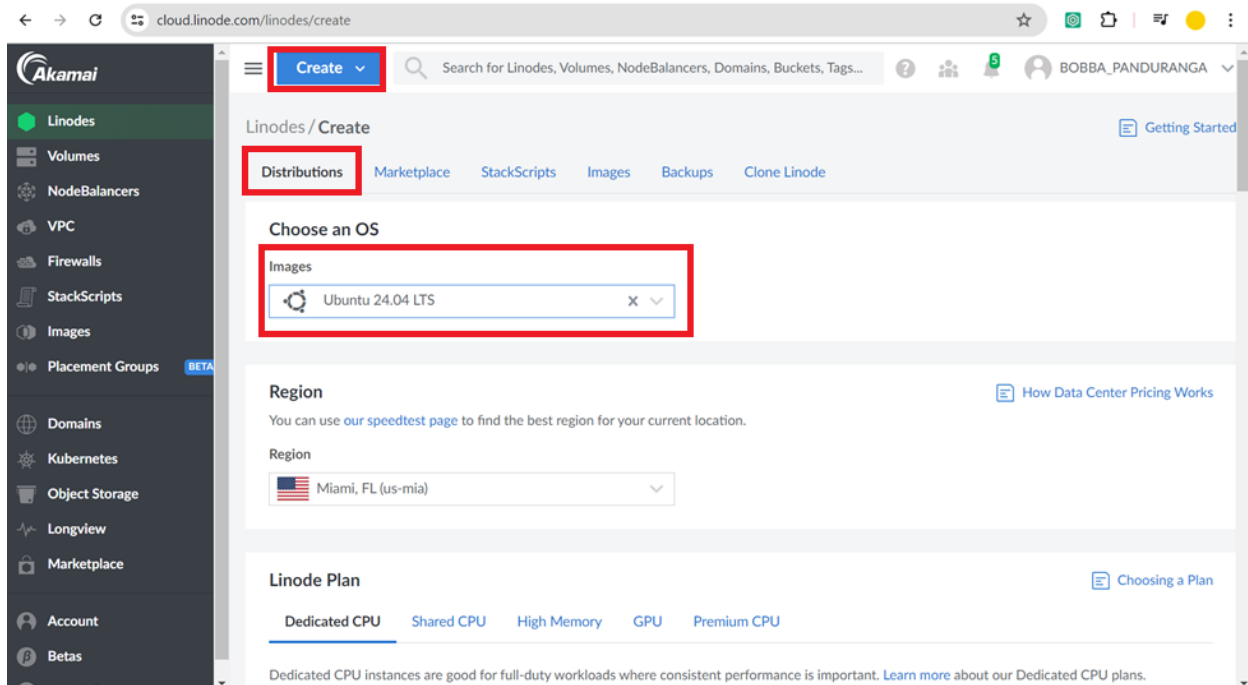
Check statistics index pattern ✓



Creating the dock server:

STEP 1: Create the new server by following the below commands

- Click on the create
- Select the distributions option
- Select the os imager as I shown below it must be Ubuntu 24.04.LTS



STEP 2: :Select The Shared CPU And Select The 2gb Version And Click On Create Linode

cloud.linode.com/linodes/create

Shared CPU | Dedicated CPU | High Memory | GPU | Premium CPU

Shared CPU instances are good for medium-duty workloads and are a good mix of performance, resources, and price. [Learn more](#) about our Shared CPU plans.

Plan	Monthly	Hourly	RAM	CPUs	Storage	Transfer	Network In / Out
☐ Nanode 1 GB	\$5	\$0.0075	1 GB	1	25 GB	1 TB	40 Gbps / 1 Gbps
☒ Linode 2 GB	\$12	\$0.018	2 GB	1	50 GB	2 TB	40 Gbps / 2 Gbps
☐ Linode 4 GB	\$24	\$0.036	4 GB	2	80 GB	4 TB	40 Gbps / 4 Gbps
☐ Linode 8 GB	\$48	\$0.072	8 GB	4	160 GB	5 TB	40 Gbps / 5 Gbps
☐ Linode 16 GB	\$96	\$0.144	16 GB	6	320 GB	8 TB	40 Gbps / 6 Gbps
☐ Linode 32 GB	\$192	\$0.288	32 GB	8	640 GB	16 TB	40 Gbps / 7 Gbps
☐ Linode 64 GB	\$384	\$0.576	64 GB	16	1280 GB	20 TB	40 Gbps / 9 Gbps
☐ Linode 96 GB	\$576	\$0.864	96 GB	20	1920 GB	20 TB	40 Gbps / 10 Gbps
☐ Linode 128 GB	\$768	\$1.152	128 GB	24	2560 GB	20 TB	40 Gbps / 11 Gbps
☐ Linode 192 GB	\$1152	\$1.728	192 GB	32	3840 GB	20 TB	40 Gbps / 12 Gbps

STEP 3: Now select the docker and open it

cloud.linode.com/linodes

Create | Search for Linodes, Volumes, NodeBalancers, Domains, Buckets, Tags... | BOBBA_PANDURANGA

Linodes | Docs | **Create Linode**

Label	Status	Plan	Public IP Address	Region	Last Backup	
docker	Running	Linode 2 GB	172.232.110.92	Chennai, IN	Never	...
pandu	Running	Linode 4 GB	172.236.74.89	Chennai, IN	Never	...

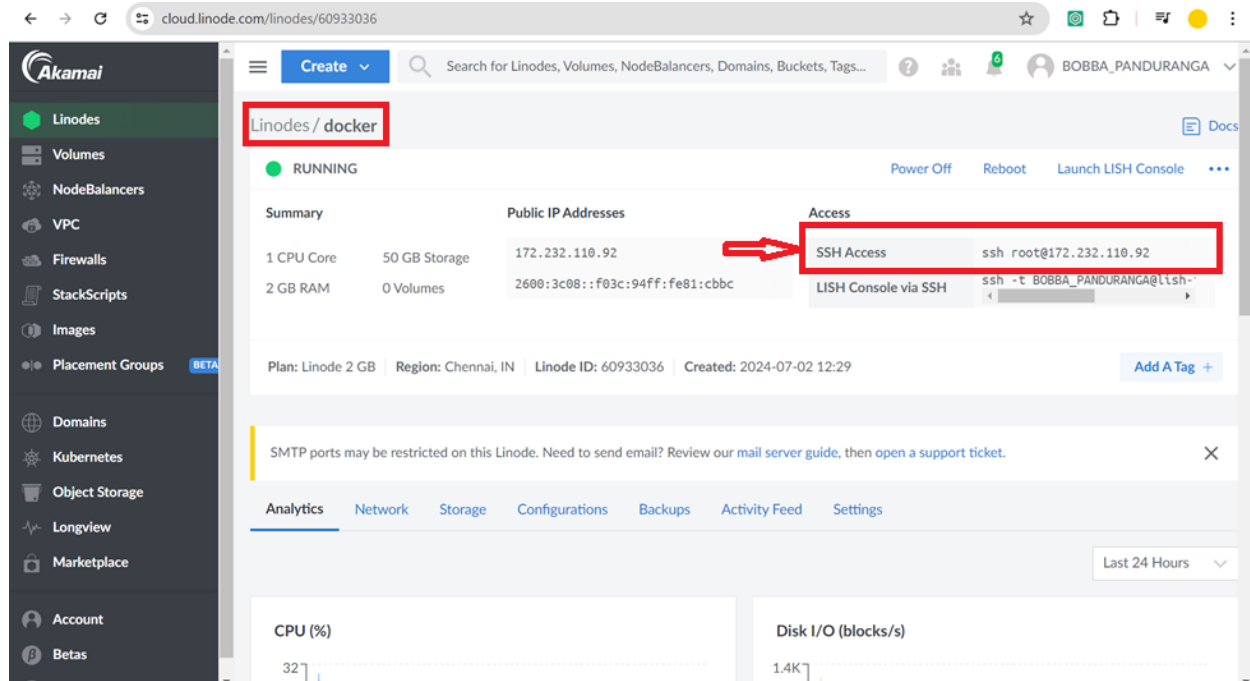
Monthly Network Transfer Pool usage: 0.02% Global Transfer Pool

[Download CSV](#)

v1.122.0 | API Reference | Provide Feedback

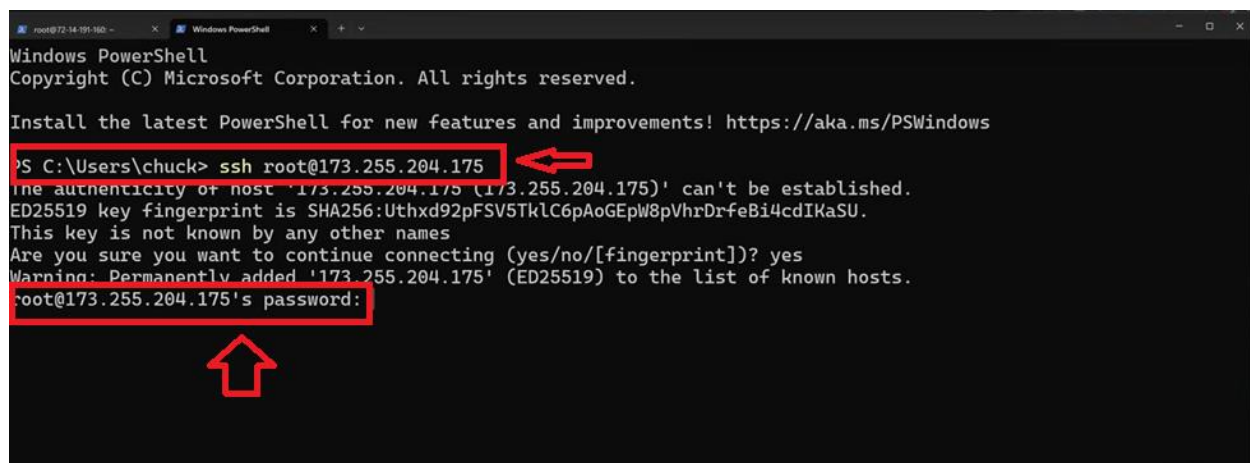
<https://cloud.linode.com/linodes/60933036>

STEP 4: Now Copy The Ssh Access



The screenshot shows the Linode dashboard for a Linode instance named 'docker'. The instance is in a 'RUNNING' state. The 'Access' tab is selected, showing the 'SSH Access' method with the command 'ssh root@172.232.110.92'. A red box highlights the 'SSH Access' section, and a red arrow points to the IP address '172.232.110.92'.

STEP 5: Past It In Cmd (Regular Mode) Enter The Password As Well



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\chuck> ssh root@173.255.204.175
The authenticity of host '173.255.204.175 (173.255.204.175)' can't be established.
ED25519 key fingerprint is SHA256:Uthxd92pFSV5TkLC6pAoGEpW8pVhrDrfeBi4cdIKaSU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '173.255.204.175' (ED25519) to the list of known hosts.
root@173.255.204.175's password:
```

STEP 6: Now create the the docker with the following commands

```
root@localhost:~#  
root@localhost:~#  
root@localhost:~#  
root@localhost:~#  
root@localhost:~# sudo apt update  
Get:1 http://mirrors.linode.com/ubuntu jammy InRelease  
Get:2 http://mirrors.linode.com/ubuntu jammy-updates InRelease  
Get:3 http://mirrors.linode.com/ubuntu jammy-backports InRelease  
0% [Connecting to security.ubuntu.com]
```

❖ Must visit this link to learn more about :

➤ <https://documentation.wazuh.com/current/deployment-options/docker/wazuh-container.html>

STEP 7: Type The Following Command To Install Docker In Your Server

```
root@localhost:~#  
root@localhost:~#  
root@localhost:~# sudo apt install docker.io docker-compose -y
```

↑
Type to install Docker

STEP 8 : Clone The Docker By Typing The Clone Command As Shown Below

```
root@localhost:~#  
root@localhost:~#  
root@localhost:~# git clone https://github.com/wazuh/wazuh-docker.git -b v4.4.5  
Cloning into 'wazuh-docker'...  
remote: Enumerating objects: 10586, done.  
remote: Counting objects: 100% (626/626), done.  
remote: Compressing objects: 100% (337/337), done.  
remote: Total 10586 (delta 320), reused 547 (delta 263), pack-reused 9960  
Receiving objects: 100% (10586/10586), 313.86 MiB | 43.15 MiB/s, done.  
Resolving deltas: 100% (5397/5397), done.  
Note: switching to 'ce8dd29425e4bc03135b5f2b71d7425b7d865f5d'.  
  
You are in 'detached HEAD' state. You can look around, make experimental  
changes and commit them, and you can discard any commits you make in this  
state without impacting any branches by switching back to a branch.  
  
If you want to create a new branch to retain commits you create, you may  
do so (now or later) by using -c with the switch command. Example:  
  
git switch -c <new-branch-name>  
  
Or undo this operation with:  
  
git switch -  
  
Turn off this advice by setting config variable advice.detachedHead to false  
  
root@localhost:~# ls  
wazuh-docker  
root@localhost:~#
```

```
root@localhost:~# ls  
wazuh-docker  
root@localhost:~# cd wazuh-docker/  
root@localhost:~/wazuh-docker# ls  
build-docker-images CHANGELOG.md indexer-certs-creator LICENSE multi-node README.md single-node VERSION  
root@localhost:~/wazuh-docker# cd single-node/  
root@localhost:~/wazuh-docker/single-node# docker-compose -f generate-indexer-certs.yml run --rm generator  
  
Digest: sha256:73cd153b2d533c4b7c4a52bb6411a8bbd03eae95f692da61e6953e4555c1526  
Status: Downloaded newer image for wazuh/wazuh-dashboard:4.4.5  
Creating single-node_wazuh.indexer_1 ... done  
Creating single-node_wazuh.manager_1 ... done  
Creating single-node_wazuh.dashboard_1 ... done  
root@localhost:~/wazuh-docker/single-node# docker stats
```

```

root@localhost: ~
CONTAINER ID   NAME                                CPU %     MEM USAGE / LIMIT   MEM %     NET I/O       BLOCK I/O
d345e358352d   single-node_wazuh.dashboard_1     0.20%     155.5MiB / 1.918GiB  7.92%      251kB / 264kB   186MB / 67.4M
B 12
0f1f6e63ed87   single-node_wazuh.indexer_1       0.86%     1.031GiB / 1.918GiB  53.76%     296kB / 244kB   1.66GB / 730M
B 65
6dc0e36b697c   single-node_wazuh.manager_1       0.80%     257.5MiB / 1.918GiB  13.11%     58.6kB / 85.7kB 421MB / 87.1M
B 155

```

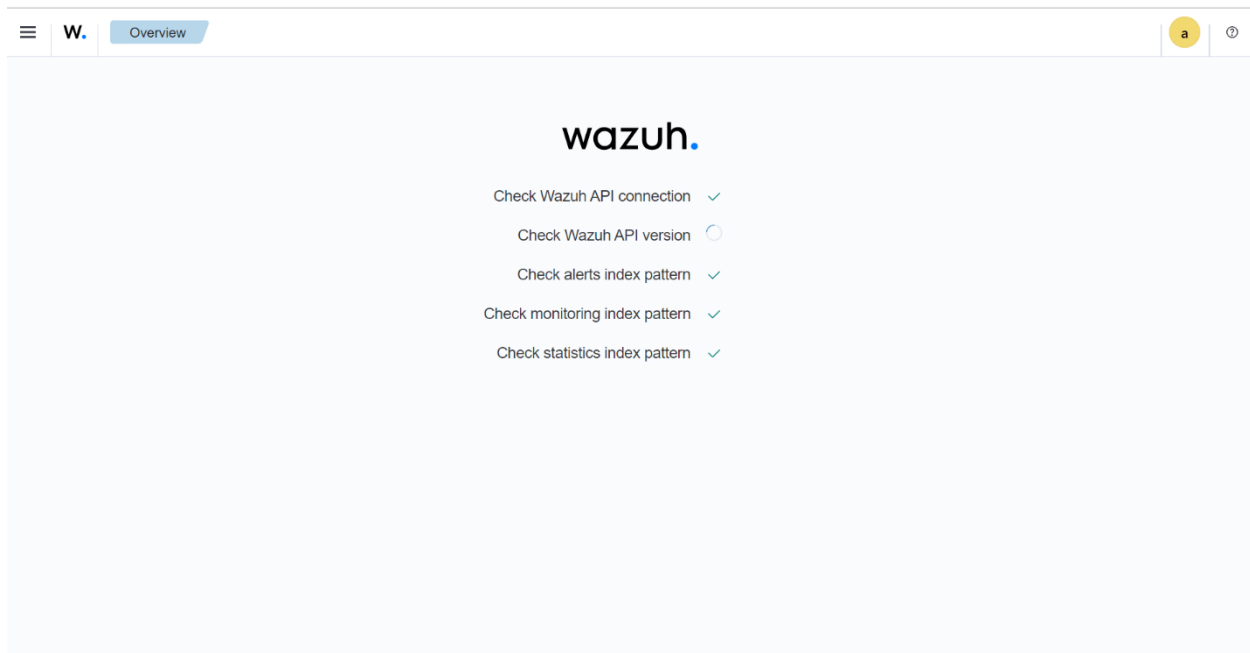
STEP 9: Now Done The Docker Is Installed In Server Now Copy The <Ip> In New Tab As https://<Ip>

The screenshot shows the Linode Cloud Manager dashboard. On the left is a navigation sidebar with options like Linodes, Volumes, NodeBalancers, VPC, Firewalls, StackScripts, Images, Placement Groups, Domains, Kubernetes, Object Storage, Longview, Marketplace, Account, Betas, and Help & Support. The main area has tabs for Analytics, Network, Storage, Configurations, Backups, Activity Feed, and Settings. Under the 'Network' tab, there are three sections:

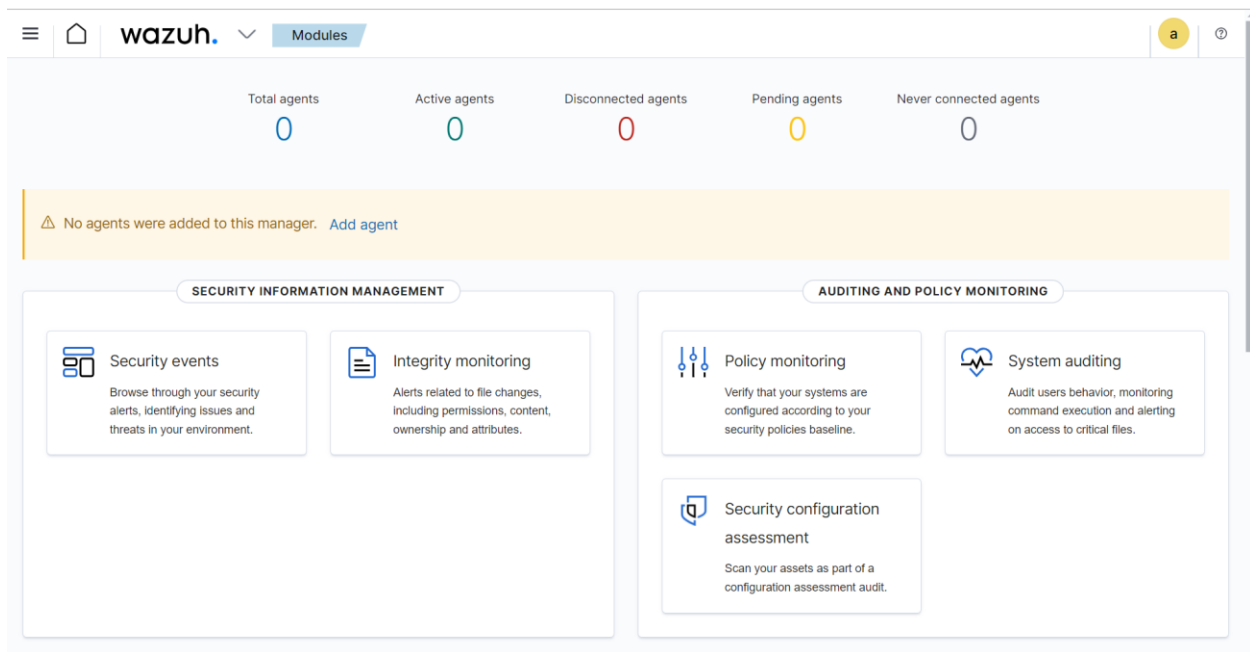
- Monthly Network Transfer:** A bar chart showing usage for July. It indicates 'docker' used 0.15 GB (1%), 'Global Pool Used' 1 GB (1%), and 'Global Pool Remaining' 5749 GB.
- Network Transfer History (Kb/s):** A line graph titled '< Last 30 Days >' showing a sharp spike in traffic around July 3rd.
- DNS Resolvers:** A list of DNS resolver IPs and their corresponding hostnames, such as 172.232.96.17 pointing to 2600:3c08:f03c:93ff:fe7c:1135.

Below these are sections for 'Firewalls' (showing no firewalls assigned) and 'IP Addresses'. In the 'IP Addresses' section, a table lists various IP addresses. One IPv4 address, 172.232.110.92, is highlighted with a red box, and its reverse DNS entry, 172-232-110-92.ip.linodeusercontent.com, is also highlighted with a red box. A red arrow points from this box down to a text instruction at the bottom of the page.

copy and past it in url bar on new window as https://<id>



Now We Have To Add The Pc To Our Sever

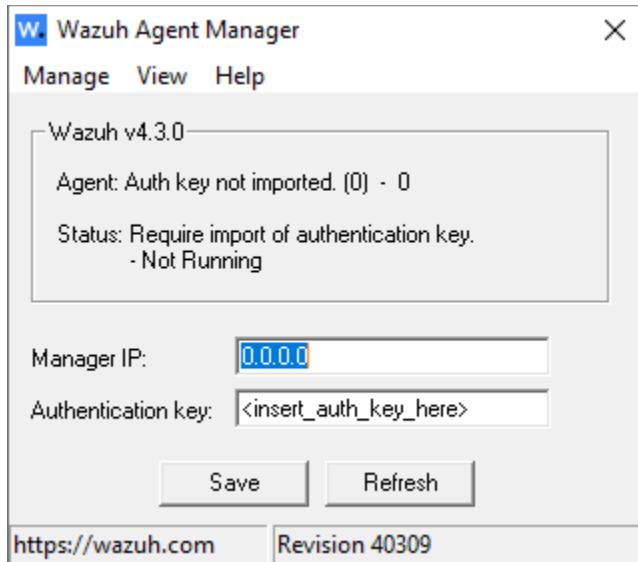


STEP 1: Download The Wazuh Agent From Below Link

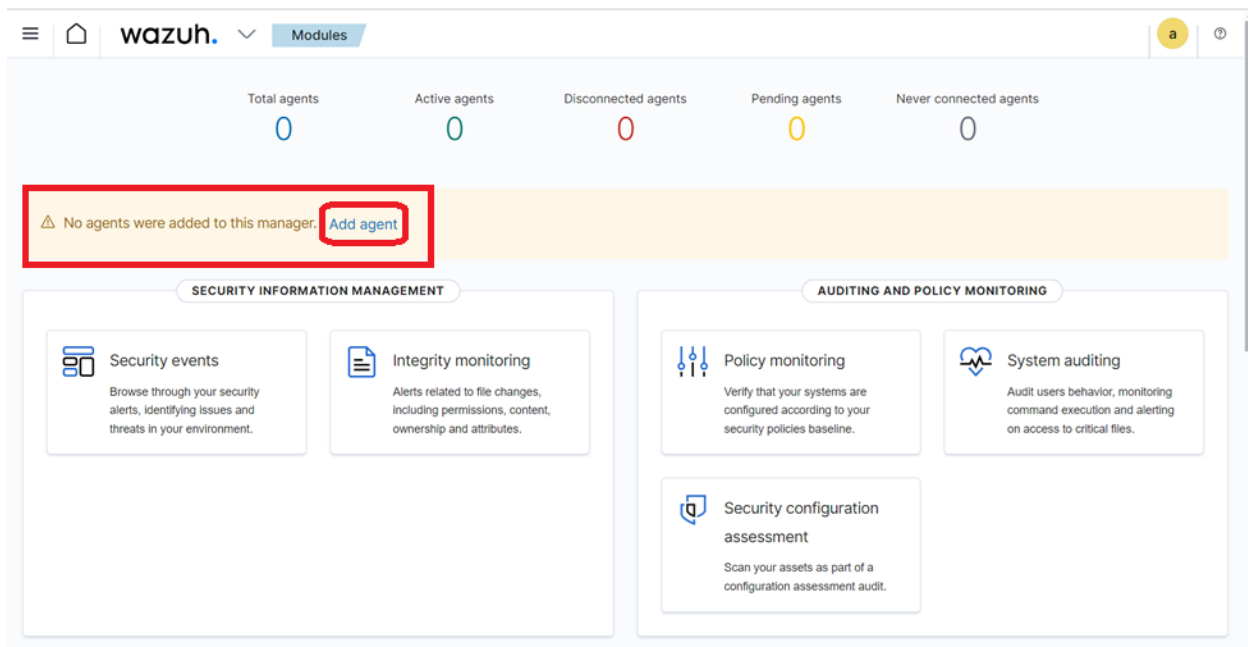
<https://documentation.wazuh.com/4.7/installation-guide/wazuh-agent/index.html>

STEP 2: Install The Wazuh-Agent And Type The IP That You Have Pasted In URL Bar

STEP 3: After Click On Save And Go To Manage Option And Click On Start Option



STEP 4: Now Click On The Agent Option



STEP 5: Enter The IP As You Enter In URL Bar

STEP 6: Copy The Command As You Got And Go To Windows Powershell (Admin) And Copy Them

STEP 7: Click On The Close Button As Shown Below

Deploy new agent

Close

1

Select the package to download and install on your system:

LINUX

☐ RPM amd64

☐ RPM aarch64

☐ DEB amd64

☐ DEB aarch64

WINDOWS

☒ MSI 32/64 bits

macOS

☐ Intel

☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

2

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

10.0.0.198

Remember server address

3

Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name

Windows

The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups

default

4

Run the following commands to download and install the agent:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.8.0-1.msi -OutFile $(env:tmp)\wazuh-agent; msixexec.exe /1 $(env:tmp)\wazuh-agent /q WAZUH_MANAGER="10.0.0.198" WAZUH_AGENT_GROUP="default" WAZUH_AGENT_NAME="Windows"
```

Requirements

- You will need administrator privileges to perform this installation.
- PowerShell 3.0 or greater is required.

Keep in mind you need to run this command in a Windows PowerShell terminal.

5

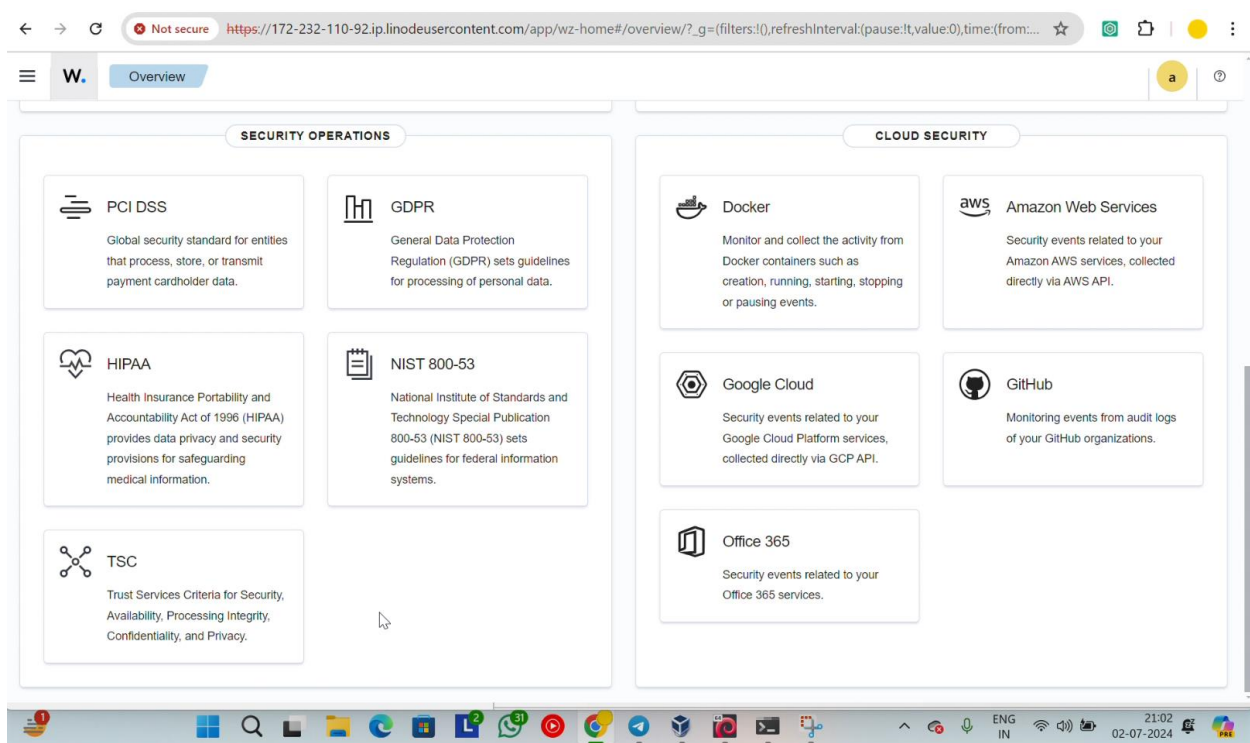
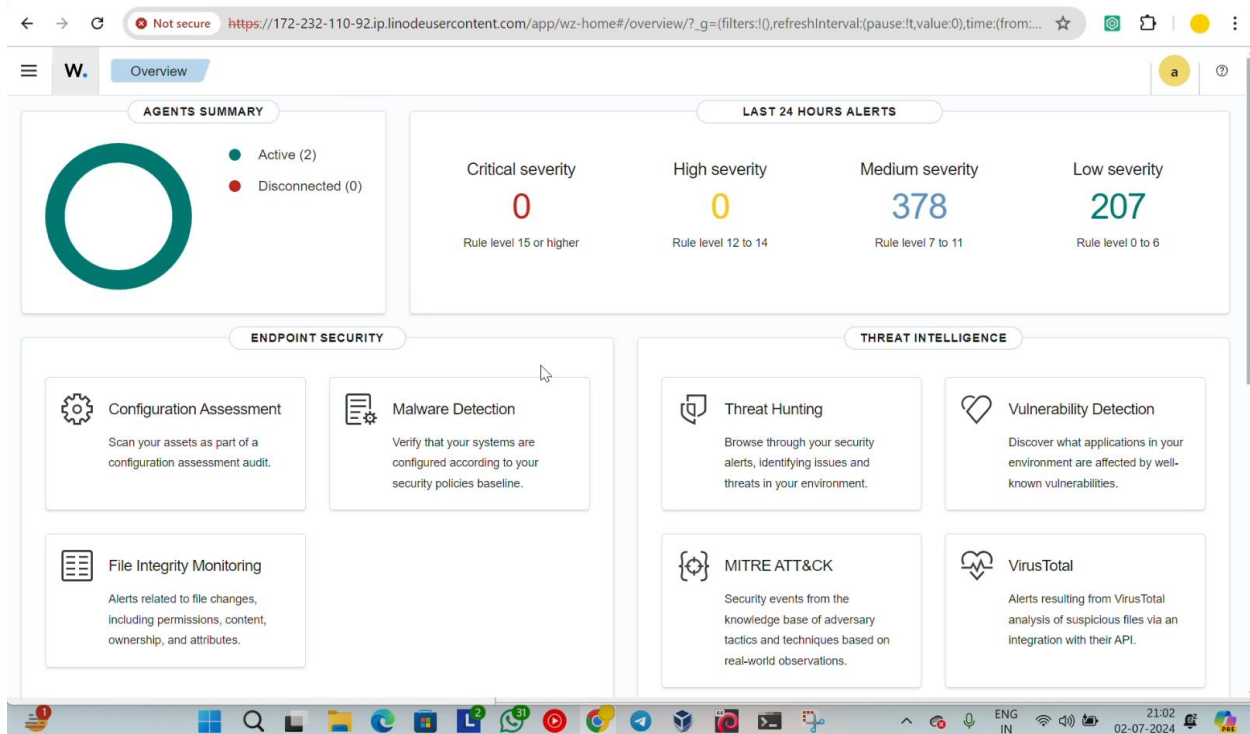
Start the agent:

```
NET START WazuhSvc
```

Close

29

STEP 8: Now You Will See Your Dashboard



STEP 9: click on the Active

The screenshot shows the Wazuh dashboard Overview page. The 'AGENTS SUMMARY' section has a red box around the 'Active (2)' status. The 'LAST 24 HOURS ALERTS' section shows 0 Critical, 0 High, 378 Medium, and 207 Low severity alerts. The 'ENDPOINT SECURITY' and 'THREAT INTELLIGENCE' sections are also visible.

AGENTS SUMMARY

- Active (2)
- Disconnected (0)

LAST 24 HOURS ALERTS

Critical severity	High severity	Medium severity	Low severity
0	0	378	207
Rule level 15 or higher	Rule level 12 to 14	Rule level 7 to 11	Rule level 0 to 6

ENDPOINT SECURITY

- Configuration Assessment
- Malware Detection
- File Integrity Monitoring

THREAT INTELLIGENCE

- Threat Hunting
- Vulnerability Detection
- MITRE ATT&CK
- VirusTotal

Endpoints

STATUS

- Active (2)
- Disconnected (0)
- Pending (0)
- Never connected (0)

DETAILS

Active	Disconnected	Pending	Never connected
2	0	0	0

Agents coverage: 100.00%

Last enrolled agent: DESKTOP-0A1S0GU

Most active agent: DESKTOP-0A1S0GU

Evolution

Last 24 hours

Agents (2)

status=active

WQL Refresh

ID	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
001	kali	10.0.2.15	default	Kali GNU/Linux 2024.1	node01	v4.8.0	active	
002	DESKTOP-0A1S0GU	10.0.2.15	default	Microsoft Windows 10 Pro 10.0.19045.3803	node01	v4.8.0	active	

Rows per page: 10

2. Malware detection [Permalink to this headline](#)

Malware, short for malicious software, refers to any software specifically designed to harm or exploit computer systems, networks, or users. It is created with the intention of gaining unauthorized access, causing damage, stealing sensitive information, or performing other malicious activities on a target system. There are various types of malware, each with specific functions and infection methods. Some common types of malware include viruses, worms, ransomware, botnets, spyware, trojans, and rootkits.

Malware detection is crucial for safeguarding computer systems and networks from cyber threats. It helps identify and mitigate malicious software that can cause a data breach, system compromise, and financial loss.

Wazuh for malware detection [Permalink to this headline](#)

Traditional methods, which rely solely on signature-based detections, have limitations and fail to capture new threats. Signature-based approaches struggle with detecting zero-day attacks, polymorphic malware, and other evasion techniques employed by threat actors. As a result, organizations are at risk of undetected breaches and data exfiltration. Wazuh empowers organizations to detect and respond to sophisticated and evasive threats effectively. Wazuh encompasses different modules that identify malware properties, activities, network connections, and more.

Detecting malicious activities with threat detection rules [Permalink to this headline](#)

Wazuh has threat detection rules that enable behavior-based malware detection. Instead of relying solely on predefined signatures, Wazuh focuses on monitoring and analyzing abnormal behavior exhibited by malware. This allows Wazuh to detect known and previously unknown threats. This way, Wazuh provides a proactive and adaptable defense

against cyber threats. Wazuh has out-of-the-box rulesets that are specifically designed to trigger alerts for recognized malware patterns, providing a quick response to potential security incidents. For example, the image below shows an alert with rule ID 92213 triggered when an executable is dropped in a folder commonly used by malware. This alert prompts security teams to begin the investigation and remediation process.

The screenshot displays the Wazuh Threat Hunting interface. On the left, a sidebar lists various fields for filtering. The main panel shows an alert for rule ID 92213, titled 'Executable file dropped in folder commonly used by malware'. Below the alert, an 'Expanded document' is shown in a table format. The table contains the following data:

Field	Value
_index	wazuh-alerts-4.x-2024.05.03
agent.id	001
agent.ip	192.168.0.104
agent.name	Windows11
data.win.eventdata.creationUtcTime	2024-05-03 15:00:50.257
data.win.eventdata.image	C:\Program Files (x86)\Microsoft\EdgeWebView\Application\114.0.1823.82\msedgeview2.exe
data.win.eventdata.processGuid	{430d8aff-0fa2-64c9-3601-000000000000}
data.win.eventdata.processId	1668
data.win.eventdata.ruleName	technique_id=T1047,technique_name=File System Permissions Weakness
data.win.eventdata.targetFileName	C:\Users\TheCotik\AppData\Local\Temp\chrome_component_unpacker_BeginUnzipping1668_1275834351\manifest.json

Wazuh allows users to create **custom rules** for more flexibility in detection, empowering them to focus on relevant activities, and optimizing malware detection. Wazuh decodes and organizes logs from monitored endpoints into fields, which can then be utilized to create custom rules for alerting when malicious activity is detected.

Wazuh rules use multiple fields that denote indicators of compromise (IOCs) to reduce false positives and detect known malware based on specific behaviors. These rules can connect related malware activities, such as intrusion, privilege escalation, lateral movement, obfuscation, and exfiltration for comprehensive detection.

Below is an example of some Wazuh custom rules created to alert on malicious activities of the LimeRAT malware:

```

<group name="lime_rat,sysmon,">

  <!-- Rogue create netflix.exe creation -->
  <rule id="100024" level="12">
    <if_sid>61613</if_sid>
    <field name="win.eventdata.image" type="pcr2">\.exe</field>
    <field name="win.eventdata.targetFilename" type="pcr2">(?!)[c-
z]:\\Users\\*.+\\AppData\\Roaming\\checker netflix\.exe</field>
    <description>Potential LimeRAT activity detected: checker netflix.exe created at
$(win.eventdata.targetFilename) by $(win.eventdata.image).</description>
    <mitre>
      <id>T1036</id>
    </mitre>
  </rule>

  <!-- Registry key creation for persistence -->
  <rule id="100025" level="12">
    <if_group>sysmon</if_group>
    <field name="win.eventdata.details" type="pcr2">(?!)[c-
z]:\\Users\\*.+\\AppData\\Roaming\\checker netflix\.exe</field>
    <field name="win.eventdata.targetObject" type="pcr2"
>HKU\\*.+\\Software\\Microsoft\\Windows\\CurrentVersion\\Run\\checker
netflix\.exe</field>
    <field name="win.eventdata.eventType" type="pcr2" >^SetValue$</field>
    <description>Potential LimeRAT activity detected: $(win.eventdata.details) added itself to the
Registry as a startup program $(win.eventdata.targetObject) to establish
persistence.</description>
    <mitre>
      <id>T1547.001</id>
    </mitre>
  </rule>

  <!-- Network activity detection -->
  <rule id="100026" level="12">
    <if_sid>61605</if_sid>

```

```

    <field name="win.eventdata.image" type="pcre2">(?!)[c-
z]:\\\\Users\\\\.+\\\\AppData\\\\Roaming\\\\checker netflix\\.exe</field>
    <description>Potential LimeRAT activity detected: Suspicious DNS query made by
$(win.eventdata.image).</description>
    <mitre>
    <id>T1572</id>
    </mitre>
</rule>

<!-- LimeRAT service creation -->
<rule id="100028" level="12">
    <if_sid>61614</if_sid>
    <field name="win.eventdata.targetObject" type="pcre2"
>HKLM\\\\System\\\\CurrentControlSet\\\\Services\\\\disk</field>
    <field name="win.eventdata.eventType" type="pcre2" >^CreateKey$</field>
    <description>Potential LimeRAT activity detected: LimeRAT service
$(win.eventdata.targetObject) has been created on $(win.system.computer).</description>
    <mitre>
    <id>T1543.003</id>
    </mitre>
</rule>

</group>

```

These rules create alerts that are visible in the **Threat Hunting** module on the Wazuh dashboard.

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
May 4, 2024 @ 21:31:34.468	001	Windows11			Potential LimeRAT activity detected: Suspicious DNS query made by C:\Users\...\AppData\Roaming\checker netfix.exe	12	100026
May 4, 2024 @ 21:31:33.462	001	Windows11			Potential LimeRAT activity detected: LimeRAT service HKLM\System\CurrentControlSet\Services\disk has been created on WinDev2210Eval	12	100028
May 4, 2024 @ 21:31:33.453	001	Windows11			Potential LimeRAT activity detected: DLL C:\Windows\Microsoft.NET\Framework\v4.0.30319\mscorlib.dll is injected on WinDev2210Eval	12	100027
May 4, 2024 @ 21:31:32.385	001	Windows11			Potential LimeRAT activity detected: C:\Users\User\AppData\Roaming\checker netfix.exe added itself to the registry HKU\{5-1-5-21-121717721-1469895683-4281812179-1000}\Software\Microsoft\Windows\CurrentVersion\Run\checker netfix.exe as a startup program to establish persistence.	12	100025
May 4, 2024 @ 21:31:31.158	001	Windows11			Potential LimeRAT activity detected: checker netfix.exe created at C:\Users\...\AppData\Roaming\checker netfix.exe by C:\Users\User\Downloads\limerat.exe.	12	100024

Refer to the blog post on [LimeRat detection and response with Wazuh](#) for the full configuration.

Wazuh identifies behavior indicative of malware, it generates real-time alerts and notifications, enabling security teams to respond swiftly and mitigate potential risks before they escalate.

Leveraging file integrity monitoring for detecting malware activity[Permalink to this headline](#)

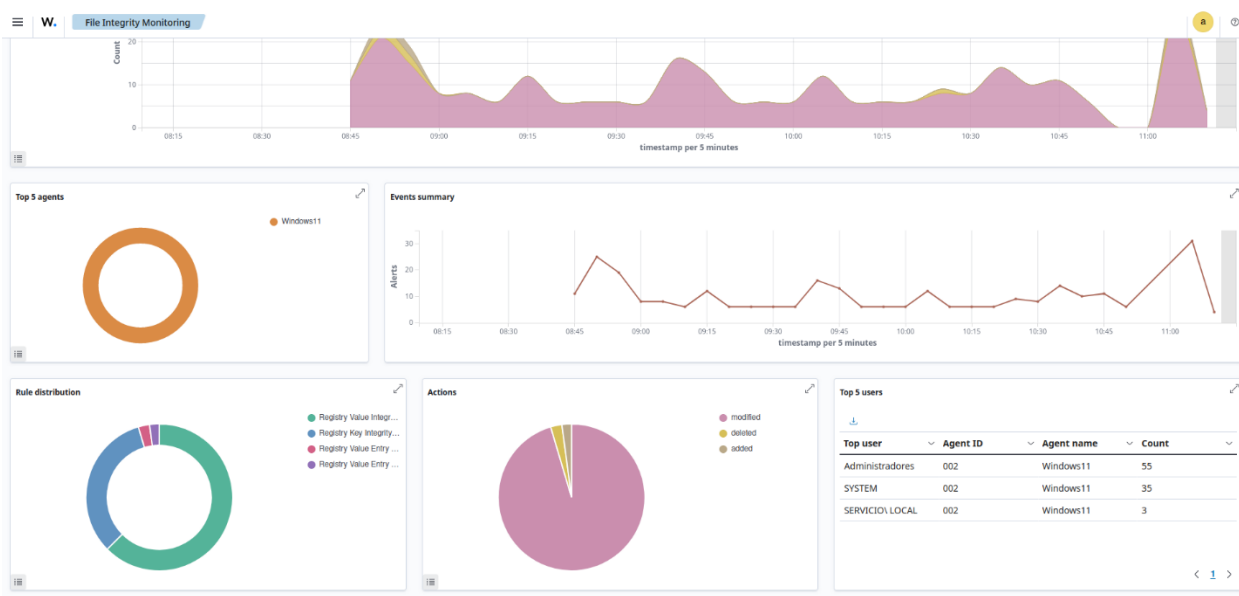
File Integrity Monitoring (FIM) is a valuable component in malware detection. Wazuh provides [FIM capabilities](#) to monitor and detect changes to files and directories on monitored endpoints. These changes include creation, modification, or deletion. While FIM provides essential insights, combining it with other capabilities and integrations further enhances its effectiveness for malware detection. Wazuh allows security teams to create custom rules based on FIM events, enabling targeted malware detection. These customizable rules correlate FIM events with specific indicators of compromises such as suspicious file extensions, code snippets, or known malware signatures.

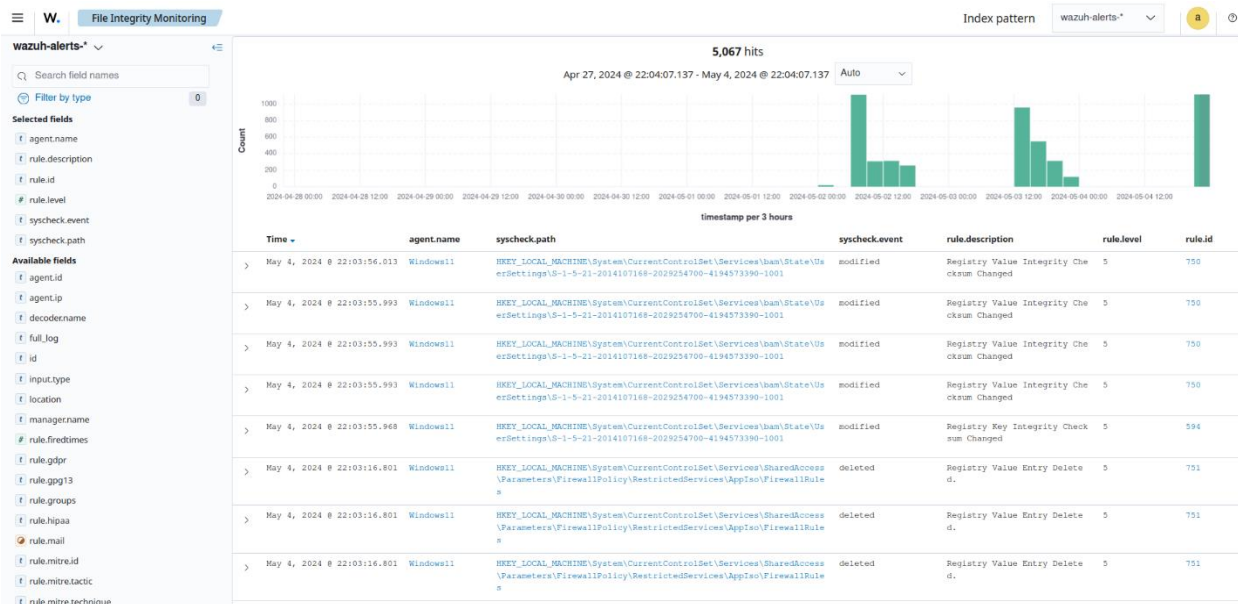
The image below shows an alert when a web shell creates or modifies a file on a web server.



Malware frequently targets Windows Registry to achieve malicious objectives, such as establishing persistence and performing other malicious actions. The Wazuh File Integrity Monitoring (FIM) module includes [Windows Registry monitoring](#) that monitors commonly targeted registry paths to detect modifications. When changes occur, the FIM module triggers real-time alerts, empowering security teams to swiftly identify and respond to suspicious registry key manipulation.

The images below display the Wazuh FIM module dashboard and events of Windows Registry modifications.





Unveiling stealthy threats with rootkit detection [Permalink to this headline](#)

Rootkits are malicious software designed to conceal the presence of malware on an endpoint by manipulating operating system functions such as altering system calls or modifying kernel data structures. Wazuh has a [Rootcheck module](#) that periodically scans the monitored endpoint to detect rootkits both at the kernel and the user space level. The rootcheck identifies and alerts potential rootkit activity. By analyzing system behavior and comparing it to known rootkit patterns, Wazuh promptly detects rootkit-related patterns and raises alerts for further investigation.

Below, we show an example of an alert generated by the Wazuh Rootcheck module when it detects an anomaly in the filesystem:

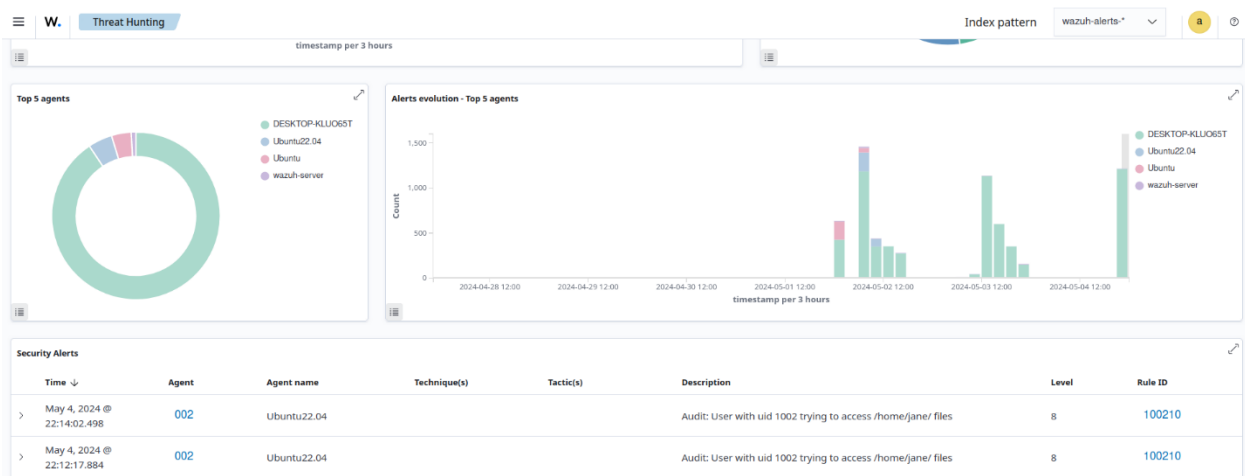
```
** Alert 1668497750.1838326: - ossec,rootcheck,pci_dss_10.6.1,gdpr_IV_35.7.d,
2022 Nov 15 09:35:50 (Ubuntu) any->rootcheck
Rule: 510 (level 7) -> 'Host-based anomaly detection event (rootcheck).'
Rootkit 't0rn' detected by the presence of file '/usr/bin/.t0rn'.
title: Rootkit 't0rn' detected by the presence of file '/usr/bin/.t0rn'.
```

While Wazuh continues to enhance its [rootkit behavior detection capabilities](#), the [Command monitoring module](#) can also be configured to monitor command-line activities across endpoints, enabling the detection of malicious commands and malware activities. This module provides organizations with a comprehensive approach to uncovering hidden threats and safeguarding their systems effectively.

Monitoring system calls for malware and anomaly detection[Permalink to this headline](#)

Wazuh [monitors system calls](#) on Linux endpoints to bolster malware detection and aid in anomaly detection. Wazuh utilizes the Linux Audit system to monitor system calls.

System call monitoring in combination with Wazuh File Integrity Monitoring (FIM) and threat intelligence integration enhances malware detection. It captures security-relevant events like file access, command execution, and privilege escalation, providing real-time insights into potential security incidents. This comprehensive approach strengthens organizations' cybersecurity resilience. In the image below, you can visualize the alerts for privilege abuse on the Wazuh dashboard for Ubuntu Linux 22.04.



Wazuh empowers security teams to leverage the audit rules provided by Auditd. Creating custom rules based on system call events enhances malware detection efforts and strengthens overall cybersecurity resilience.

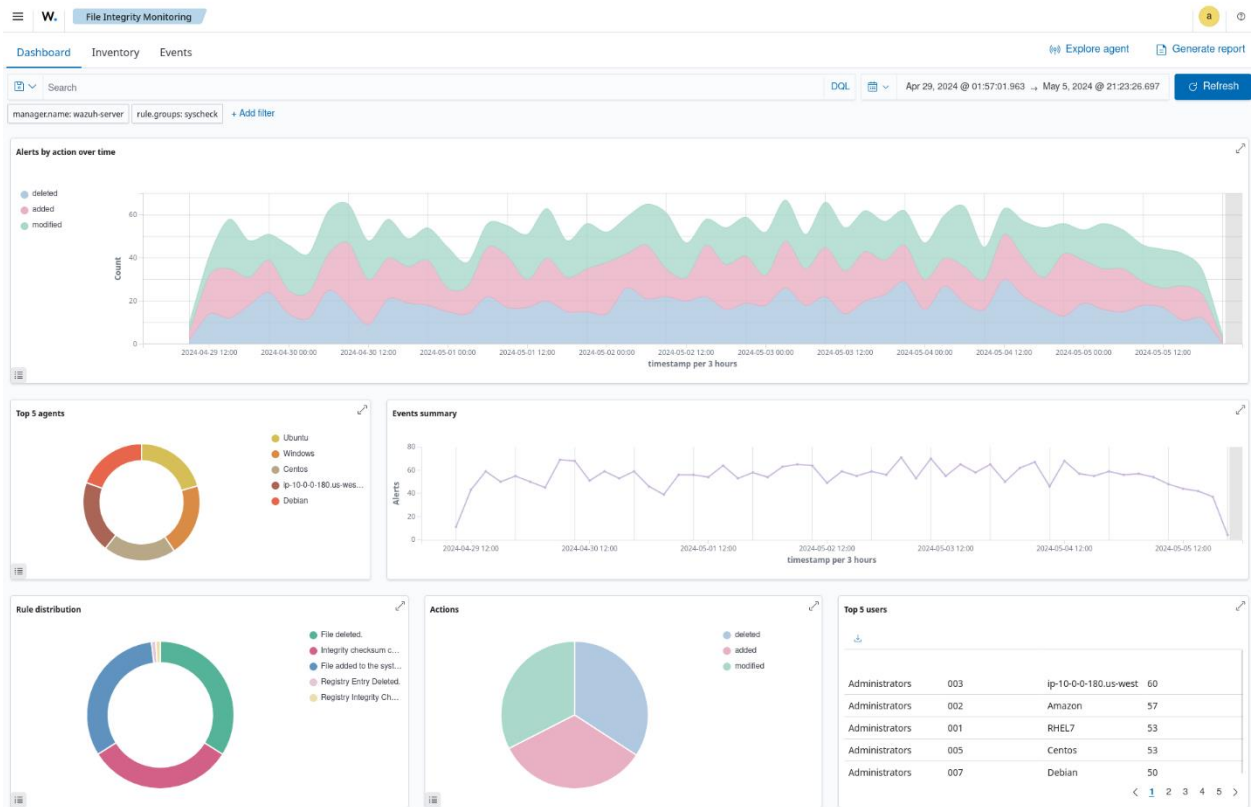
File Integrity Monitoring

You can find a dedicated **File Integrity Monitoring** module in the Wazuh dashboard where all file integrity events triggered from monitored endpoints are reported. This increases visibility as it provides valuable information on the status of monitored directories and their potential impact on the security posture. The **Wazuh FIM dashboard** has three different sections to view FIM analysis results; **Inventory, Dashboard, and Events**.

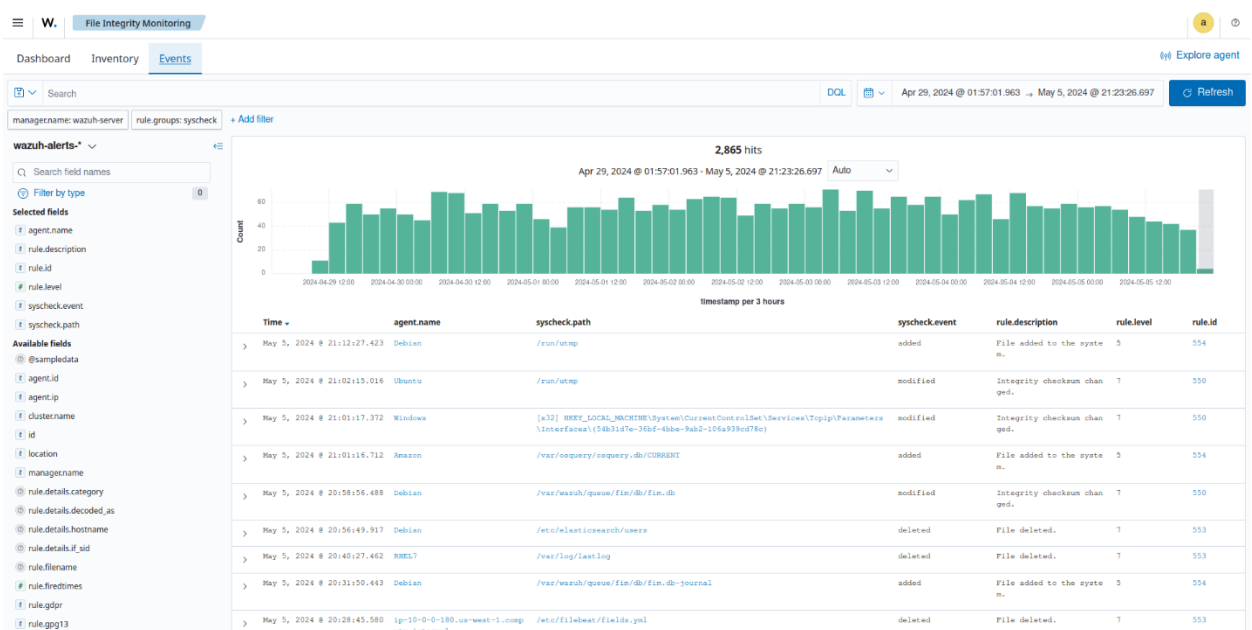
1. The **Inventory** section displays a list of all files that the FIM module has indexed. Each file has entry information including the filename, last modification date, user, user ID, group, and file size.

File	Last Modified	User	User ID	Group	Group ID	Size
/bin	Apr 15, 2024 @ 23:02:50.000	root	0	root	0	7
/etc/passwd.lock	Apr 15, 2024 @ 23:02:56.000	root	0	root	0	0
/etc/adduser.conf	Apr 15, 2024 @ 23:02:57.000	root	0	root	0	3028
/etc/alternatives/README	Apr 6, 2022 @ 06:40:25.000	root	0	root	0	100
/etc/alternatives/awk	Apr 15, 2024 @ 23:05:39.000	root	0	root	0	13
/etc/alternatives/nawk	Apr 15, 2024 @ 23:05:39.000	root	0	root	0	13
/etc/alternatives/pager	Apr 15, 2024 @ 23:06:02.000	root	0	root	0	9
/etc/alternatives/rmt	Apr 15, 2024 @ 23:05:57.000	root	0	root	0	17
/etc/alternatives/which	Apr 15, 2024 @ 23:03:16.000	root	0	root	0	26
/etc/apt/apt.conf.d/01-vendor-ubuntu	Apr 8, 2022 @ 07:22:23.000	root	0	root	0	92
/etc/apt/apt.conf.d/01-autoremove	Apr 8, 2022 @ 07:22:23.000	root	0	root	0	630
/etc/apt/apt.conf.d/70debconf	Feb 20, 2022 @ 11:42:49.000	root	0	root	0	182
/etc/apt/apt.conf.d/docker-autoremove-suggests	Apr 15, 2024 @ 23:06:21.000	root	0	root	0	44
/etc/apt/apt.conf.d/docker-clean	Apr 15, 2024 @ 23:06:21.000	root	0	root	0	318
/etc/apt/apt.conf.d/docker-disable-periodic-update	Apr 15, 2024 @ 23:06:21.000	root	0	root	0	27

2. The **Dashboard** section shows an overview of the events triggered by the FIM module for all monitored endpoints. You can also streamline it to show the events for a selected monitored endpoint.



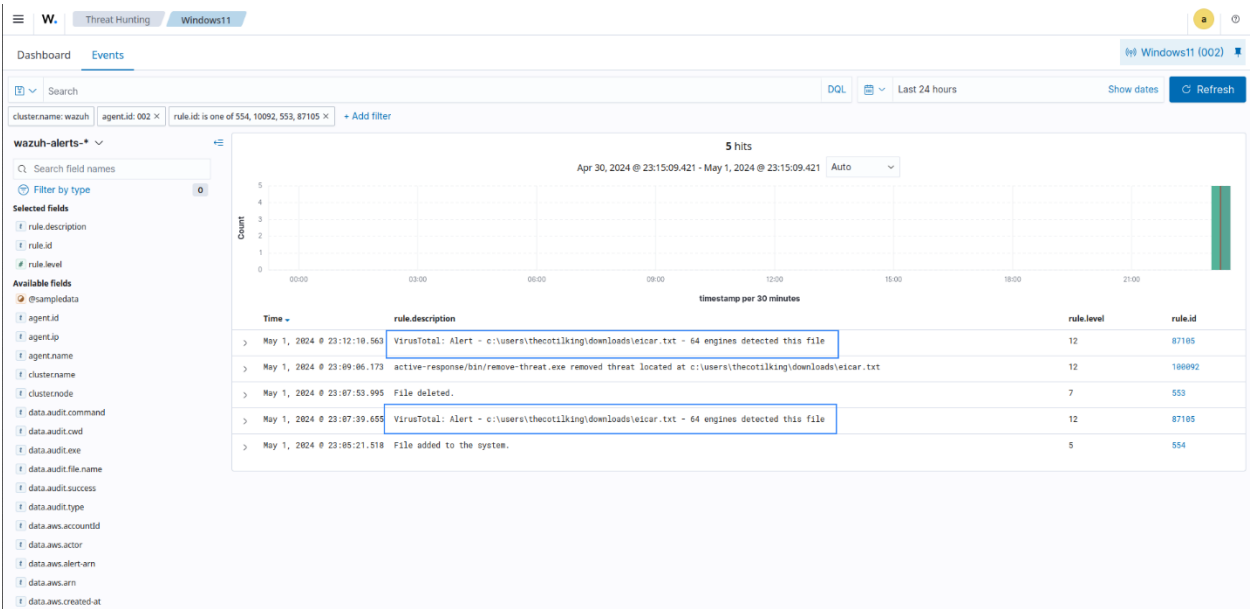
- The **Events** section shows the alerts triggered by the FIM module. It displays details such as the agent name, the file path of the monitored file, the type of FIM event, a description of the alert, and the rule level of each alert.



Below are common use cases the Wazuh FIM module would assist you in monitoring within your environment.

Enhancing malware detection with threat intelligence integration

Users can boost their malware detection capabilities by [integrating with threat intelligence](#) sources. These intelligence feeds enrich the Wazuh knowledge base with additional up-to-date information on known malicious IP addresses, domains, URLs, and other indicators of compromise. Examples of threat intelligence sources Wazuh can integrate with include VirusTotal, MISP, and more.

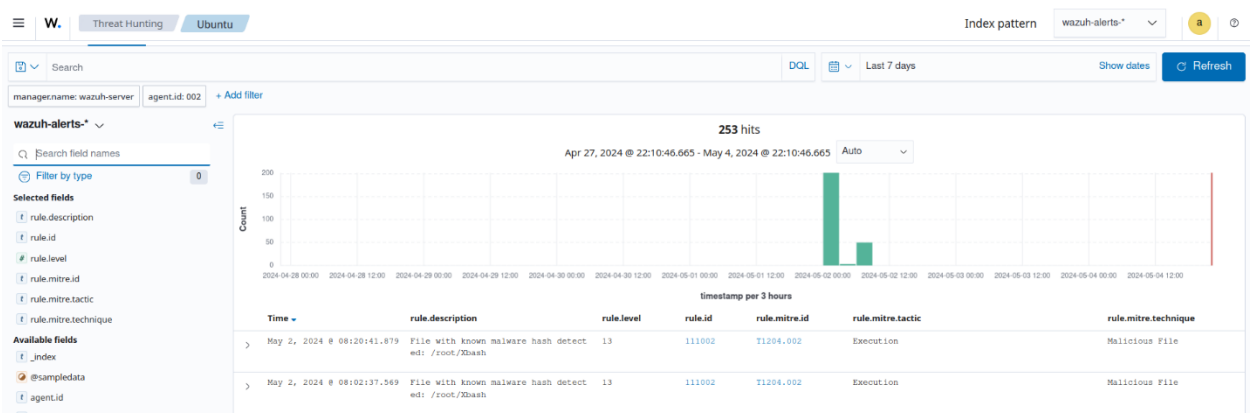


Wazuh proactively identifies malicious files by comparing the identified IOCs with the information stored in the [CDB lists](#) (constant databases). These lists can store known malware indicators of compromise (IOCs) including file hashes, IP addresses, and domain names.

You can customize entries in either `key:value` or `key:` format for tailored detection, an example of such is seen below. A CBD list containing known MD5 malware hashes of the Mirai and Xbash malware is used for detection:

```
e0ec2cd43f71c80d42cd7b0f17802c73:mirai
55142f1d393c5ba7405239f232a6c059:Xbash
```

Upon detection, these alerts are observed within the **Threat Hunting** module of the Wazuh dashboard, as seen below.



Refer to the [Use case: Detecting malware using file hashes in a CDB list](#) for full configurations.

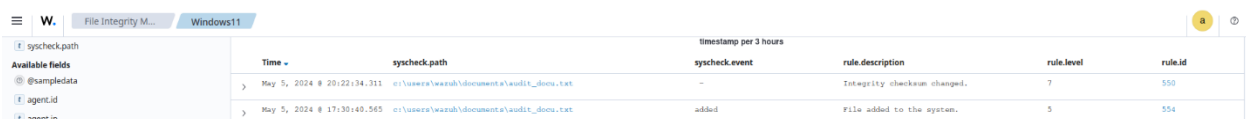
Monitoring file integrity[Permalink to this headline](#)

Modifications to configuration files and file attributes are frequent occurrences within endpoints in an IT infrastructure. However, if not validated, there may be unauthorized and inadvertent changes that could affect the behavior of the endpoints or the applications running in them. The Wazuh FIM module runs periodic scans on specific files and directories to detect file changes in real time. It scans the designated files to create a baseline of the current state. It checks for file modifications by comparing checksums and attribute values to the baseline, generating alerts if discrepancies are found.

The Wazuh FIM module supports various configuration options that enable effective monitoring of assets:

- **Real-time monitoring:** The FIM module provides a **realtime** attribute that enables continuous monitoring of specified directories. This feature is particularly useful for monitoring critical directories and tracking changes immediately after they occur. Wazuh allows you to specify the directories or files in the monitored endpoints that would be reported in real-time if file changes occur.
- **Scheduled monitoring:** The **frequency** option in the Wazuh FIM module allows users to customize the scheduling of each FIM scan performed in your monitored endpoints. The default scan interval for the FIM module is 12 hours (43200 seconds) and can be customized on each endpoint. Alternatively, scans can be scheduled using the **scan_time** and the **scan_day** options. These options help users to set up FIM scans outside business hours or during holidays.
- **Who-data monitoring:** Wazuh captures advanced insights into file changes using the who-data functionality. This functionality uses audit tools like the Linux Audit subsystem and Microsoft Windows SACL to determine important information about the detected file changes. The who-data monitoring functionality allows the FIM module to obtain information on when the change event occurred, who or what made the change, and what content was changed. This is useful in maintaining accountability and validating if changes made to monitored files or directories were authorized and performed using approved processes.

Below is an example of an alert generated when a monitored file is changed on a Windows endpoint.

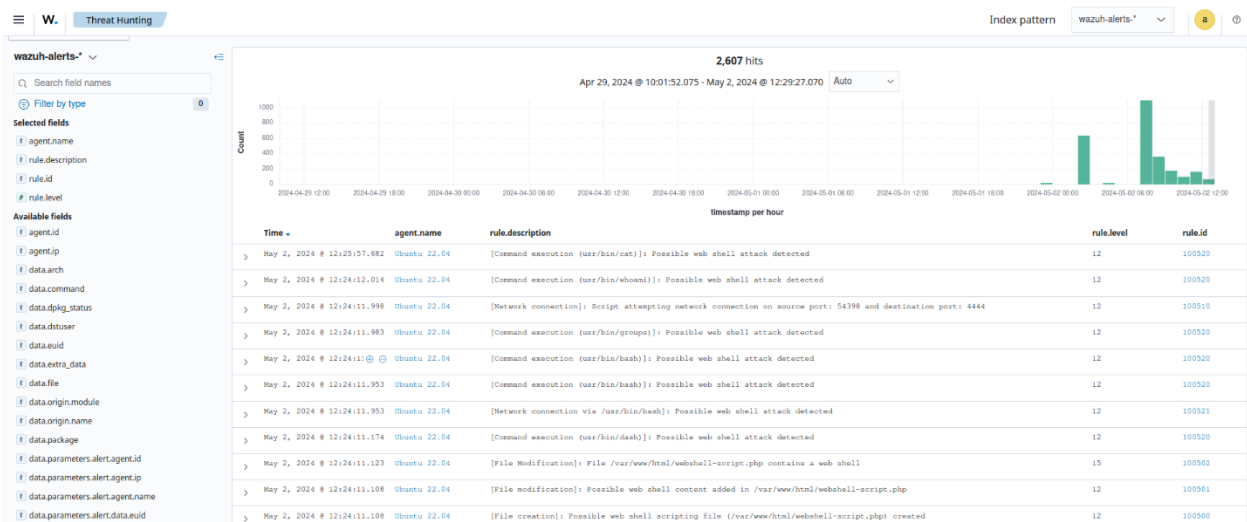


Time	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
May 5, 2024 @ 20:22:34.311	c:\users\wazuh\documents\audit_docu.txt	-	Integrity checksum changed.	7	550
May 5, 2024 @ 17:30:40.565	c:\users\wazuh\documents\audit_docu.txt	added	File added to the system.	5	554

In alert fields, the *who-data* metadata shows that the user **wazuh** added the word **Hello** to the **audit_docu.txt** file using the **Notepad.exe** process.



- **Reporting changes in file values:** The FIM module provides a **report_changes** attribute that records and reports the exact content changed in a text file to the Wazuh server. The attribute enables the Wazuh agent to make copies of monitored files to a private location on each endpoint for further review. This monitoring option is helpful when users want to initiate specific responses when file changes in monitored directories match the behavior of known malicious activities. For example, the alert below indicates when Wazuh detects the creation of a web shell scripting file **webshell-script.php** in a monitored directory.



- **Recording file attributes:** Users can configure the FIM module to record specific attributes of a monitored file. Wazuh supports various file attributes that users can

use to specify the file metadata that the FIM module will record or ignore. For example, this monitoring option would be useful when users want to record only the SHA-256 hash of a configuration file, excluding other hash types.

Detecting and responding to malware[Permalink to this headline](#)

The Wazuh FIM module integrates with other Wazuh capabilities and third-party threat intelligence solutions to create a comprehensive security monitoring environment. This is imperative to enhance malware detection and response capabilities, ensuring robust defense against cyber threats.

The Wazuh FIM module supports various integrations, including but not limited to:

- **File integrity monitoring and YARA:** By combining the Wazuh FIM module and the YARA tool, it is possible to detect malware when suspicious file additions or modifications are identified. The YARA rule files contain samples of malware indicators that are downloaded to the monitored endpoints. When the FIM module detects a change in the monitored file or directory, it executes a YARA scan using a script to determine if it is malware. If the YARA rule finds a match with a file, it will send the scan results to the Wazuh server for decoding and alerting. This would be reported according to the custom rule and decoder configurations configured on the Wazuh server. Check this documentation for more information on [how to integrate the Wazuh FIM module with YARA](#).
- **File integrity monitoring and VirusTotal:** The Wazuh [Integrator module](#) connects to external APIs and alerting tools such as VirusTotal. The [VirusTotal integration](#) uses the VirusTotal API to detect malicious file hashes within the files and directories monitored by the FIM module. Once enabled, when FIM generates alerts, Wazuh initiates the VirusTotal integration to extract the hash value associated

with the flagged file from the alert. The VirusTotal API is then used to compare these hashes against its scanning engines for potentially malicious content.

- **File integrity monitoring and active response:** The [Wazuh active response](#) module automatically responds to threats identified in a timely manner. This combination enables the FIM module to not only detect but also respond to malicious activities. You can configure active response scripts to execute when the FIM module detects file changes in your monitored environment. Additionally, it also generates alerts for the response performed. This reduces the Mean Time To Respond (MTTR) as malicious changes detected are remediated in a timely manner.

In the image below Wazuh triggers when a file is added to the monitored endpoint. The VirusTotal API scans the file and identifies it as malicious content on 55 engines. Then the Wazuh active response module acts immediately to remove the threat from the monitored endpoint.



- **File integrity monitoring and CDB list:** Wazuh FIM module also detects malicious files by checking the presence of known malware signatures when combined with [CDB lists \(constant database\)](#). CDB lists are used to store known malware indicators of Compromise (IOCs) such as file hashes, IP addresses, and domain names. When CDB lists are created, Wazuh checks if field values from FIM alerts

such as file hash match the keys stored in the CDB lists. If matched, it generates an alert and response based on how you configure your custom rule.

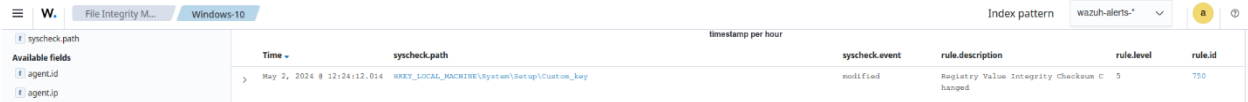


Monitoring Windows Registry[Permalink to this headline](#)

The Wazuh FIM module periodically scans Windows Registry entries, stores its checksums and attributes in a local database, and alerts when changes in registry values are detected. This would keep users informed about registry modifications resulting from user activities or software installations whether malicious or not.

You can configure the Wazuh open source FIM module to monitor [Windows Registry values](#) using various configuration options. The `report_changes` attribute in the `windows_registry` option provides a granular breakdown of modification detected in the monitored Windows Registry value. You can configure which Windows Registry attributes the module would record or ignore. For example, you can choose to record the `check_sha1sum` attribute and ignore the `check_md5sum` attribute, if your CDB list only contains SHA1 hashes of malicious files.

The image below shows the event of a modified Windows registry value in a monitored endpoint.



The alert when expanded shows the modified field.

W. File Integrity Mon... Windows-10	Index pattern: wazuh-alerts-*
- syscheck.mime_before - syscheck.mime_after - syscheck.sha1_before - syscheck.sha1_after - syscheck.sha256_before - syscheck.sha256_after - syscheck.size_before - syscheck.size_after - syscheck.tags - syscheck.uid_after - syscheck.username_after - syscheck.value_name - syscheck.value_type - syscheck.win_perm_after - timestamp	- syscheck.changed_attributes: size, md5, sha1, sha256 - syscheck.diff: diff - syscheck.event: modified - syscheck.md5_after: 0271850350a106db05310c1e65e7c - syscheck.md5_before: 47f43e746d5737dbd1e93e28f08ec4 - syscheck.mode: scheduled - syscheck.path: HKEY_LOCAL_MACHINE\System\Setup\Custom_key - syscheck.sha1_after: 7e08ac870bc8e8e7082b7c986280c2b2cf01de - syscheck.sha1_before: 336c907c10f603ba93e0eb2c50e348e740608 - syscheck.sha256_after: 92341d8e4eef92a0f3c8a4a30e4d0a70a78e903e64ac9742a806439905c - syscheck.sha256_before: 5908ecc40b44681fc9a730870241ae525ee0f703b241c3c364fa8f90b022 - syscheck.size_after: 0 - syscheck.value_name: FIN - syscheck.value_type: REG_SZ - timestamp: May 2, 2024 @ 12:24:12.014

Threat actors maintain persistence by commonly adding programs for their malicious activities to the *Run* and *RunOnce* keys in the Registry. Additionally, Wazuh detects any suspicious programs added to the startup registry keys. This allows you to take appropriate action to remove them before they cause harm to your system.

W.

File Integrity M...

Windows-10

Index pattern

wazuh-alerts-*

syscheck.path

Available fields

agent.id

agent.ip

agent.name

decoder.name

full_log

id

input.type

location

timestamp per hour

Time

May 2, 2024 @ 12:24:11.967

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnce

added

Registry Value Entry Added to the System

5

752

May 2, 2024 @ 12:24:11.953

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnce

modified

Registry Key Integrity Checksum C changed

5

594

May 2, 2024 @ 12:24:11.078

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run

added

Registry Value Entry Added to the System

5

752

May 2, 2024 @ 12:19:32.132

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run

modified

Registry Key Integrity Checksum C changed

5

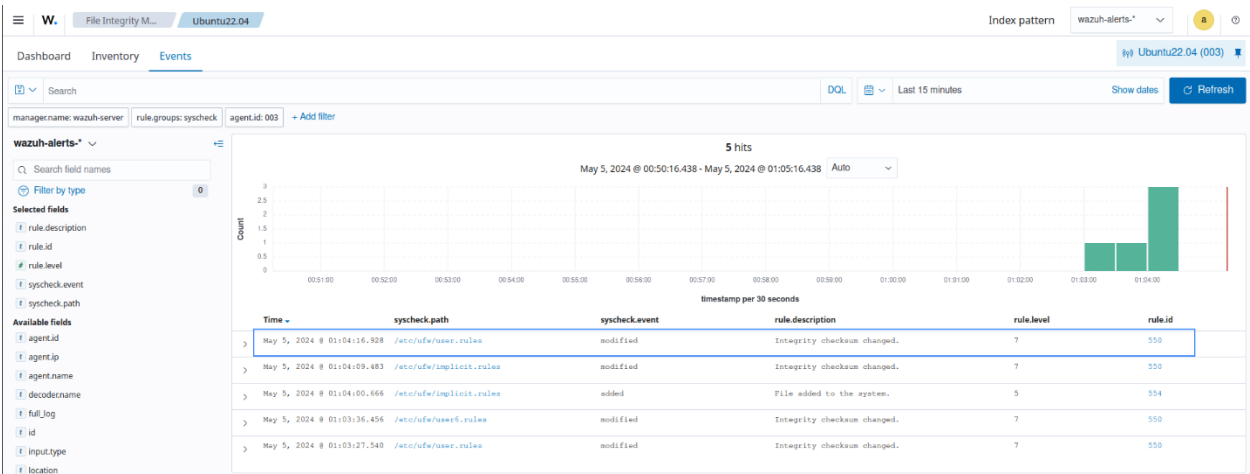
594

Meeting regulatory compliance[Permalink to this headline](#)

Meeting regulatory compliance requirements is an important consideration for organizations in various industries. File integrity monitoring is a requirement for achieving compliance with regulations such as PCI DSS, SOX, HIPAA, NIST SP 800-53, among others.

You can customize the Wazuh FIM module to monitor specific files and directories where your organization's sensitive and confidential data are stored. Wazuh provides a comprehensive report that outlines the changes made to the files and directories being monitored. This feature is particularly useful for ensuring compliance with various regulatory standards.

For example, organizations can meet the [CM-3 Configuration change control](#) requirement in NIST SP 800-53 standard by using Wazuh. The control requires organizations to protect information at rest and monitor configuration changes in their infrastructure. The image below shows an event generated when the permissions for Uncomplicated Firewall (UFW) rule files are modified on a monitored endpoint.



[Log data analysis](#)

Log data analysis is a crucial process that involves examining and extracting valuable insights from log files created by different systems, applications, or devices. These logs contain records of events that provide useful information for troubleshooting, security analysis and monitoring, and optimizing performance. Log data analysis is an essential practice that contributes to a secure, efficient, and reliable IT ecosystem.

Wazuh collects, analyzes, and stores logs from endpoints, network devices, and applications. The Wazuh agent, running on a monitored endpoint collects and forwards system and application logs to the Wazuh server for analysis. Additionally, you can send log messages to the Wazuh server via syslog or third-party API integrations.

[Log data collection](#)

Wazuh collects logs from a wide range of sources, enabling comprehensive monitoring of various aspects of your IT environment. You can check our documentation on [Log data collection](#) to understand better how Wazuh collects and analyzes logs from monitored endpoints. Some of the common log sources supported by Wazuh include:

- **Operating system logs:** Wazuh collects logs from several operating systems, including [Linux](#), [Windows](#), and [macOS](#).

Wazuh can collect syslog, auditd, application logs, and others from Linux endpoints.

Wazuh collects logs on Windows endpoints using the Windows event channel and Windows event log format. By default, the Wazuh agent monitors the System, Application, and Security Windows event channels on Windows endpoints. The Wazuh agent offers the flexibility to configure and monitor other [Windows event channels](#).

Wazuh utilizes the unified logging system (ULS) to collect logs on macOS endpoints. The macOS ULS centralizes the management and storage of logs across all the system levels.

The image below shows an event collected from the [Microsoft-Windows-Sysmon/Operational](#) event channel on a Windows endpoint.

W. Threat Hunting Windows 11

Index pattern: wazuh-alerts-*

Time: May 5, 2024 @ 12:59:02.334 rule.description: Executable dropped in Windows root folder rule.level: 6 rule.id: 92217

Expanded document

Table JSON

Field	Value
data.win.eventdata.creationTime	2024-05-05 12:59:00.215
data.win.eventdata.image	C:\Windows\System32\svchost.exe
data.win.eventdata.processGuid	{0a74b4f-3f79-64d1-7600-00000000a00}
data.win.eventdata.processId	4324
data.win.eventdata.targetFilename	C:\Windows\SoftwareDistribution\Download\install\AM_Delta.exe
data.win.eventdata.user	NT AUTHORITY\SYSTEM
data.win.eventdata.userName	NT AUTHORITY\SYSTEM
data.win.eventdata.targetDomainName	Microsoft-Windows-System/Operational
data.win.eventdata.targetLinkedLogonId	Windows11
data.win.eventdata.targetUserSid	11
data.win.eventdata.targetUserSid	0x3e7
data.win.eventdata.subjectProcessName	DESKTOP-KLU0607E
data.win.eventdata.subjectProcessId	8-1-5-18
data.win.eventdata.targetDomainName	NT AUTHORITY

- **Syslog events:** Wazuh gathers logs from [syslog-enabled](#) devices, encompassing a wide array of sources including Linux/Unix systems and network devices that do not support agent installation. The image below shows an alert triggered when a new user is created on the Linux endpoint and the log is forwarded to the Wazuh server via rsyslog.

W. Threat Hunting

Index pattern: wazuh-alerts-*

Time: May 5, 2024 @ 13:09:41.272 Agent: 000 Agent name: wazuh-server Technique(s): T1136 Tactics: Persistence Description: New user added to the system. Level: 8 Rule ID: 5902

Table JSON Rule

Field	Value
@timestamp	2024-05-05T16:09:41.272Z
_id	e2e5Y8B9PWdpLhrKSH
agent.id	000
agent.ip	192.168.0.154
agent.name	wazuh-server
data.dstuser	Stephen
data.gid	1005
data.home	/home/Stephen
data.shell	/bin/bash
data.uid	1004
decoder.name	useradd

- **Agentless monitoring:** The Wazuh agentless monitoring module monitors endpoints that don't support agent installation. It requires an SSH connection between the endpoint and the Wazuh server. The Wazuh [agentless monitoring](#) module monitors

files, directories, or configurations and runs commands on the endpoint. The image below is an alert from an agentless device on the Wazuh dashboard.

The screenshot displays the Wazuh dashboard interface. On the left, the 'Discover' tab is active, showing a search bar and a list of selected fields. The main panel shows a table of alerts with columns for Time (timestamp) and rule.description. A bar chart is visible in the background. On the right, the 'Document Details' panel is open, showing the JSON details of a specific alert. The JSON data includes fields like _index, agent.ephemeral_id, agent.hostname, agent.id, agent.name, agent.type, agent.version, agentless.host, agentless.script, agentless.user, decoder.name, full_log, id, and input.type. The full_log field is highlighted with a blue box, showing a file integrity check alert.

```

{
  "_index": "wazuh-alerts-4.x-2024.05.05",
  "agent.ephemeral_id": "6fa06291-e23d-4a30-ad9b-20f0f172ee7a",
  "agent.hostname": "wazuh",
  "agent.id": "3e201657-df0c-4c9d-8518-aa9556aaf110",
  "agent.name": "wazuh",
  "agent.type": "filebeat",
  "agent.version": "7.10.2",
  "agentless.host": "192.168.0.137",
  "agentless.script": "ssh_integrity_check_linux",
  "agentless.user": "agentless",
  "decoder.name": "syscheck_integrity_changed",
  "full_log": {
    "type": "syscheck_integrity_changed",
    "size": 1024,
    "old_md5sum": "b914e0805fda36c2f80ab8ffca65a",
    "new_md5sum": "2319e0f4b0d180912c7f61a688f6b31",
    "old_sha1sum": "a62206a7b0aa08a05f55c0e618a6e84952ae97e",
    "new_sha1sum": "0ed614d7a147335ee09de75f970e6fa06a71d"
  },
  "id": "1714930301.0300160",
  "input.type": "log"
}

```

- **Cloud provider logs:** Wazuh integrates with cloud providers like [AWS](#), [Azure](#), [Google Cloud](#), and [Office 365](#) to collect logs from cloud services such as EC2 instances, S3 buckets, Azure VMs, and more. The image below shows the **CLOUD SECURITY** section in the Wazuh dashboard.

The screenshot displays the Wazuh dashboard interface, specifically the 'Overview' tab. The dashboard is divided into several sections: ENDPOINT SECURITY, THREAT INTELLIGENCE, SECURITY OPERATIONS, and CLOUD SECURITY. The CLOUD SECURITY section is highlighted with a blue border and contains several sub-sections: Docker, AWS Amazon Web Services, Google Cloud, GitHub, and Office 365. Each sub-section provides a brief description of the security events it monitors.

- **Custom logs:** You can configure Wazuh to collect and parse logs from several applications and third-party security tools like [VirusTotal](#), [Windows Defender](#), and [ClamAV](#). The image below shows an alert of a log from VirusTotal processed by the Wazuh server.



Rules and decoders [Permalink to this headline](#)

Wazuh [rules and decoders](#) are core components in log data analysis and threat detection and response. Wazuh provides a powerful platform for log data analysis, allowing organizations to enhance their security posture by promptly detecting and responding to potential security threats.

Wazuh decoders are responsible for parsing and normalizing log data collected from various sources. Decoders are essential for converting the raw log data in several formats into a unified and structured format that Wazuh can process effectively. Wazuh has pre-built decoders for common log formats such as syslog, Windows event channel, macOS ULS, and more. Additionally, Wazuh allows you to define [custom decoders](#) for parsing logs from specific applications or devices with unique log formats. By using decoders, Wazuh can efficiently interpret log data and extract relevant information, such as timestamps, log levels, source IP addresses, user names, and more. As shown below, you can view Wazuh

out-of-the-box and custom decoders on the **Server management > Decoders** of the Wazuh dashboard.

W. Decoders

Index pattern wazuh-alerts-*

9

Decoders (1568)

From here you can manage your decoders.

Manage decoders files Add new decoders file Refresh Export formatted

Search WQL Custom decoders

Name	Program name	Order	File ↑	Path
wazuh			0005-wazuh_decoders.xml	ruleset/decoders
agent-buffer		level	0005-wazuh_decoders.xml	ruleset/decoders
agent-upgrade	agent.id, agent.name, status		0005-wazuh_decoders.xml	ruleset/decoders
agent-upgrade	error		0005-wazuh_decoders.xml	ruleset/decoders
agent-upgrade	agent.cur_version		0005-wazuh_decoders.xml	ruleset/decoders
agent-upgrade	agent.new_version		0005-wazuh_decoders.xml	ruleset/decoders
agent-restart	module		0005-wazuh_decoders.xml	ruleset/decoders
fim-state			0005-wazuh_decoders.xml	ruleset/decoders
json			0006-json_decoders.xml	ruleset/decoders
wazuh-api			0007-wazuh-api_decoders.xml	ruleset/decoders

Rows per page: 10 < 1 2 3 4 5 ... 157 >

Wazuh ruleset detects security events and anomalies in log data. These rules are written in a specific format and they trigger alerts when certain conditions are met. The rules are defined based on certain criteria like log fields, values, or patterns to match specific log entries that may indicate security threats. Wazuh provides a wide range of pre-built rules covering common security use cases. Additionally, administrators can create **custom rules** tailored to their specific environment and security requirements. The **Server management** category of the Wazuh dashboard lets you view the default and custom **Rules**.

W. Rules

Index pattern wazuh-alerts-*

9

Rules (4382)

From here you can manage your rules.

Manage rules files Add new rules file Refresh Export formatted

Search WQL Custom rules

ID ↑	Description	Groups	Regulatory compliance	Level	File	Path
1	Generic template for all syslog rules.	syslog		0	0010-rules_config.xml	ruleset/rules
2	Generic template for all firewall rules.	firewall		0	0010-rules_config.xml	ruleset/rules
3	Generic template for all ids rules.	ids		0	0010-rules_config.xml	ruleset/rules
4	Generic template for all web rules.	web-log		0	0010-rules_config.xml	ruleset/rules
5	Generic template for all web proxy rules.	squid		0	0010-rules_config.xml	ruleset/rules
6	Generic template for all windows rules.	windows		0	0010-rules_config.xml	ruleset/rules
7	Generic template for all wazuh rules.	ossec		0	0010-rules_config.xml	ruleset/rules
200	Grouping of wazuh rules.	wazuh		0	0016-wazuh_rules.xml	ruleset/rules
201	Agent event queue rule	agent_flooding, wazuh		0	0016-wazuh_rules.xml	ruleset/rules
202	Agent event queue is level full.	agent_flooding, wazuh	PCI_DSS GDPR	7	0016-wazuh_rules.xml	ruleset/rules

Rows per page: 10 < 1 2 3 4 5 ... 439 >

For example, the rule below includes a `match` field used to define the pattern that the rule looks for. The rule also has a `level` field that specifies the priority of the resulting alert. Additionally, rules enrich events with technique identifiers from the MITRE ATT&CK framework and map them to regulatory compliance controls.

```
<rule id="5715" level="3">
  <if_sid>5700</if_sid>
  <match>^Accepted|authenticated.$</match>
  <description>sshd: authentication success.</description>
  <mitre>
    <id>T1078</id>
    <id>T1021</id>
  </mitre>

  <group>authentication_success,gdpr_IV_32.2,gpg13_7.1,gpg13_7.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,pci_dss_10.2.5,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>
</rule>
```

Log data indexing and storage [Permalink to this headline](#)

The [Wazuh indexer](#) is a highly scalable, distributed real-time search and analytics engine. The Wazuh indexer is critical in log analysis as it stores and indexes alerts generated by the Wazuh server. These alerts are stored as JSON documents.

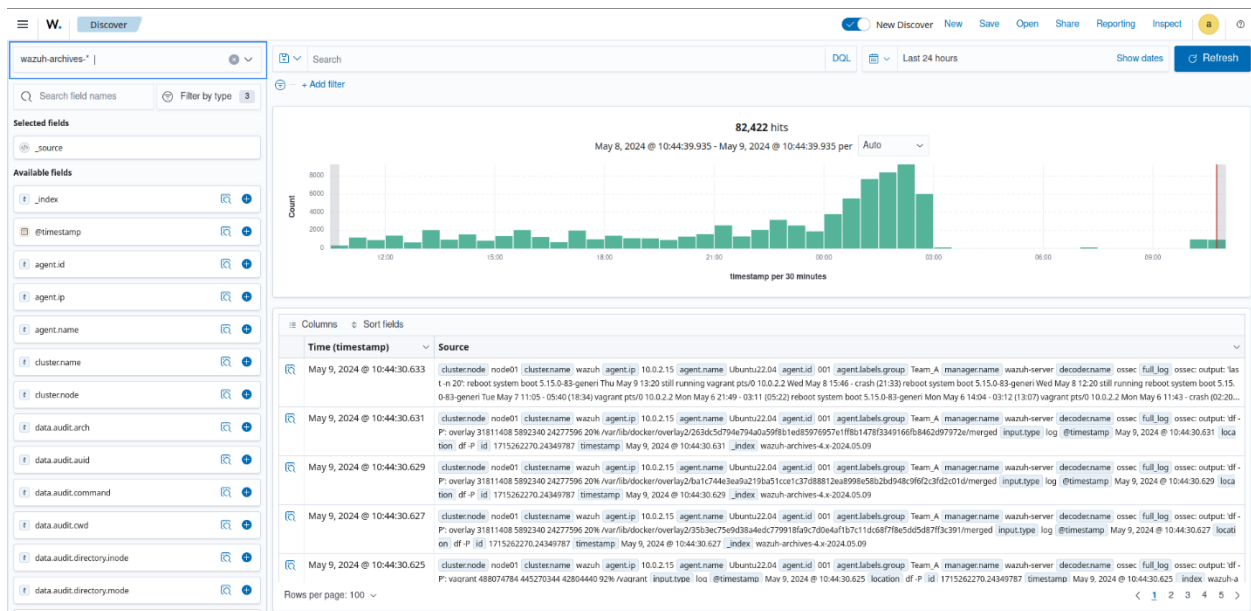
The Wazuh indexer guarantees redundancy by storing the JSON documents across several containers called shards and distributing the shards across multiple nodes. This implementation prevents downtime when hardware failures or cyber-attacks occur and increases query capacity as nodes are added to a cluster.

Wazuh uses four indices to store several event types:

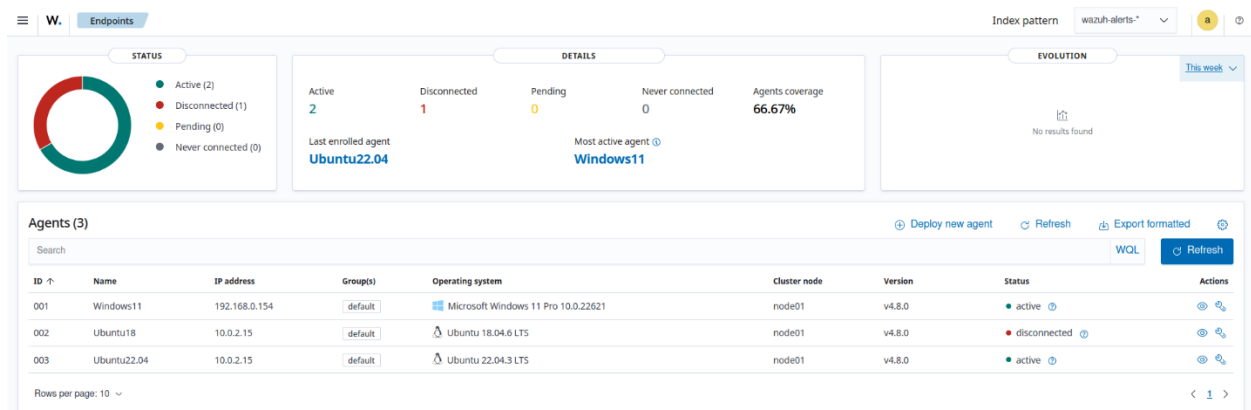
- Wazuh Discover interface showing search results for 'wazuh-alerts-*'. The interface includes a sidebar with filters for 'Selected fields' and 'Available fields'. The main area displays a bar chart titled '11,326 hits' showing the distribution of alerts over time. The x-axis is labeled 'timestamp per 30 minutes' and the y-axis is labeled 'Count'. The chart shows a significant spike in alerts around May 4, 2024, at 14:58:43.538. Below the chart, a table lists the top 5 hits with columns for 'Time (timestamp)', 'Source', and 'Log message'. The log messages are truncated, but they appear to be related to system events and user settings.

Time (timestamp)	Source	Log message
May 5, 2024 at 14:58:16.359	syscheck_value_type REG_BINARY syscheck_size_after 24 syscheck_md5_before ee5ee6af70308c672a1554a360c39 syscheck_sha256_before 0957421b3fe36a174db72c59205770549bcfa5132f7a937a4d37ae59fa syscheck_mode scheduled syscheck_path HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\lsassm\State\UserSettings\S-1-5-21-2014107168-2029254700-4194573390-1001 syscheck_sha1_after 3e3b65d5c4e760675576b6877bce4856077a9e syscheck_changed_attributes md5_sha1 sha256 syscheck_arch [x32] syscheck_event modified syscheck_md5_after d9331fed37c06972711f058c4241c262 syscheck_sha1_bef...	
May 5, 2024 at 14:58:16.353	syscheck_gid_before 5-1-5-32-544 syscheck_username_after Administradores syscheck_mtime_after May 5, 2024 at 14:58:06.000 syscheck_gid_after 5-1-5-18 syscheck_win_perm_after ('allowed': ['DELETE', 'READ_CO...	
May 5, 2024 at 14:57:27.622	syscheck_value_type REG_BINARY syscheck_size_after 24 syscheck_md5_before 5c735807b3991913d48226d6f6dd62d2 syscheck_sha256_before b3326a5a8811f6eb38616056740978ae549c464ceacfa772772c0eba7f3 syscheck_mode scheduled syscheck_path HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\lsassm\State\UserSettings\S-1-5-21-2014107168-2029254700-4194573390-1001 syscheck_sha1_after b6213665716684738273a290b191d1a1c990e2 syscheck_changed_attributes md5_sha1 sha256 syscheck_arch [x32] syscheck_event modified syscheck_md5_after ee5ee6af70308c672a1554a360c39 syscheck_sha1_bef...	
May 5, 2024 at 14:57:27.580	syscheck_gid_before 5-1-5-32-544 syscheck_username_after Administradores syscheck_mtime_after May 5, 2024 at 14:57:05.000 syscheck_gid_after 5-1-5-18 syscheck_win_perm_after ('allowed': ['DELETE', 'READ_CO...	
May 5, 2024 at 14:57:07.183	input_type log agent_ip 192.168.0.134 agent_name DESKTOP-KLU0657 agent_id 001 [management] wazuh-server data.win.eventdata.subjectLogonId 0x3e7 data.win.eventdata.subjectDomainName WORKGR...	

- 57

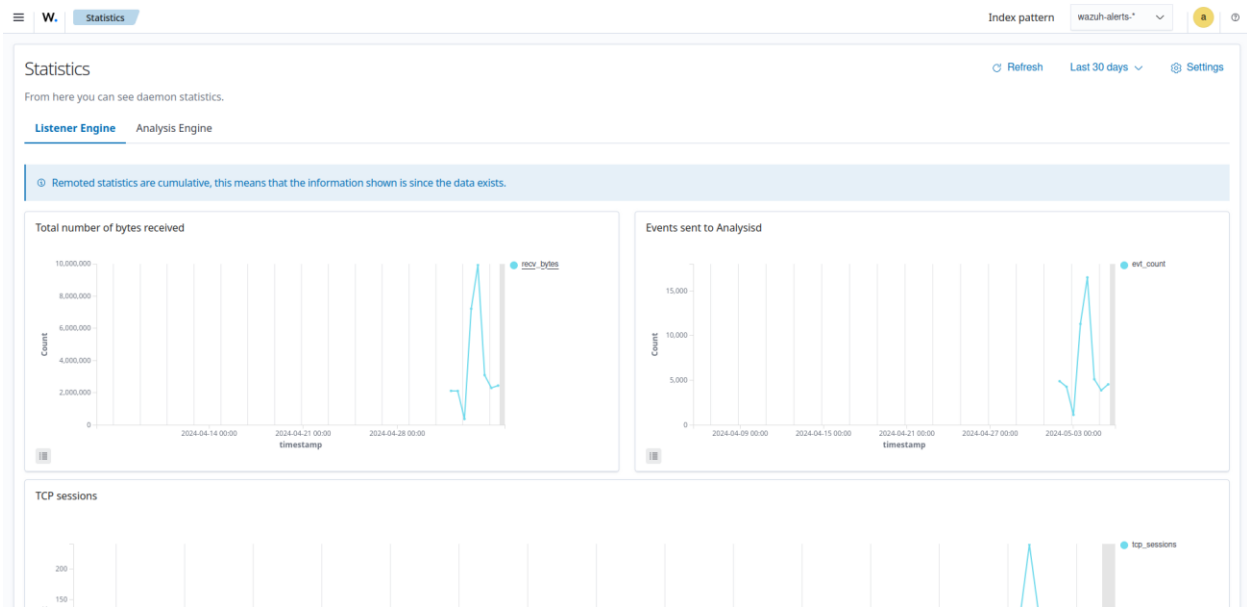


- **wazuh-monitoring** index stores data about the state of Wazuh agents over a period of time. The state of the agent could be **Active**, **Disconnected**, or **Never connected**. This information is very useful in tracking Wazuh agents that are not reporting to the dashboard for several reasons that need investigation. The image below shows the connection status of the agents on the Wazuh dashboard. The agent information as shown in the image is collected from the **wazuh-monitoring** index.



- **wazuh-statistics** index stores performance data related to the Wazuh server. This information is critical to ensuring the Wazuh server performs optimally with the

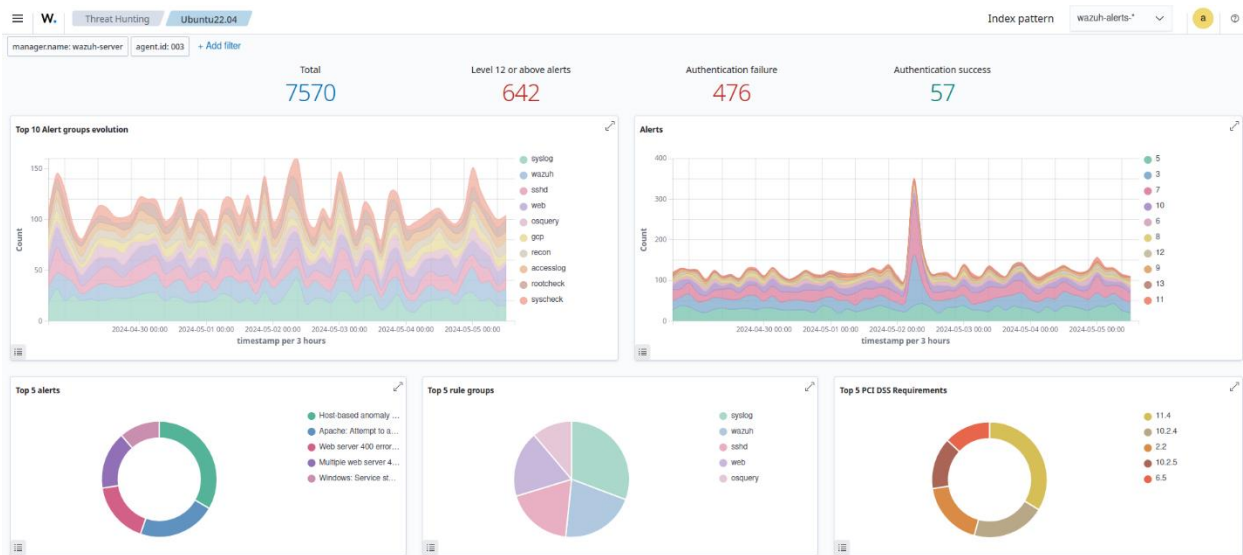
available computing resources. The image below shows performance-related events on the Wazuh dashboard.



Log data querying and visualization[Permalink to this headline](#)

The Wazuh dashboard offers log data querying and visualization capabilities. You can leverage the dashboard's intuitive interface to conduct complex searches and queries to extract meaningful insights from the log data collected by Wazuh.

Wazuh provides a set of predefined dashboards and visualizations out of the box, specifically tailored to security monitoring and compliance use cases. These dashboards provide insight into common security events such as failed logins, malware detection, and system anomalies. You can further customize these dashboards to suit your specific needs and requirements. Below is a sample image of the **Security event** dashboard showing several interesting information like **Top 5 PCI DSS Requirements**, **Top 5 alerts**, and **Alert groups evolution**.



The Wazuh dashboard enables users to explore log entries in real time, apply various filters, and drill down into specific events or time ranges. This flexibility allows security analysts to identify trends, anomalies, and potential security incidents within their environment.

Wazuh allows users to [create customized dashboards](#) that display key performance indicators, security metrics, and real-time monitoring of critical systems and applications. Users can assemble multiple visualizations, such as pie charts, line graphs, and heat maps, onto a single dashboard, providing a holistic view of their infrastructure's security posture. The following blog posts detailed how to query and create custom dashboards:

- [Monitoring macOS resources with Wazuh](#)
- [Monitoring Linux resources with Wazuh](#)
- [Monitoring Windows resources with Performance Counters](#)

[Vulnerability detection](#)[Permalink to this headline](#)

Software vulnerabilities are weaknesses in code that can allow attackers to gain access to or manipulate the behavior of an application. Vulnerable software applications are

commonly targeted by attackers to compromise endpoints and gain a persistent presence on targeted networks.

Vulnerability detection is the process of identifying these flaws before they are discovered and exploited by attackers. The goal of vulnerability detection is to identify vulnerabilities so that remediation can be carried out to prevent successful attacks.

The [Wazuh agent](#) uses the [Syscollector](#) module to collect inventory details from the monitored endpoint. It sends the collected data to the Wazuh server. Within the Wazuh server, the [Vulnerability Detection](#) module correlates the software inventory data with vulnerability content documents to detect vulnerable software on the monitored endpoint.

Wazuh detects vulnerable applications, generating risk reports, using our Cyber Threat Intelligence (CTI) platform. In this platform, we aggregate vulnerability data from diverse sources like operating system vendors and vulnerability databases, consolidating it into a unified, reliable repository. The process involves standardizing the varied formats into a common structure. Additionally, we maintain the integrity of our vulnerability data by doing the following.

- Rectifying format inconsistencies like version errors and typos.
- Completing missing information.
- Incorporating new cybersecurity vulnerabilities.

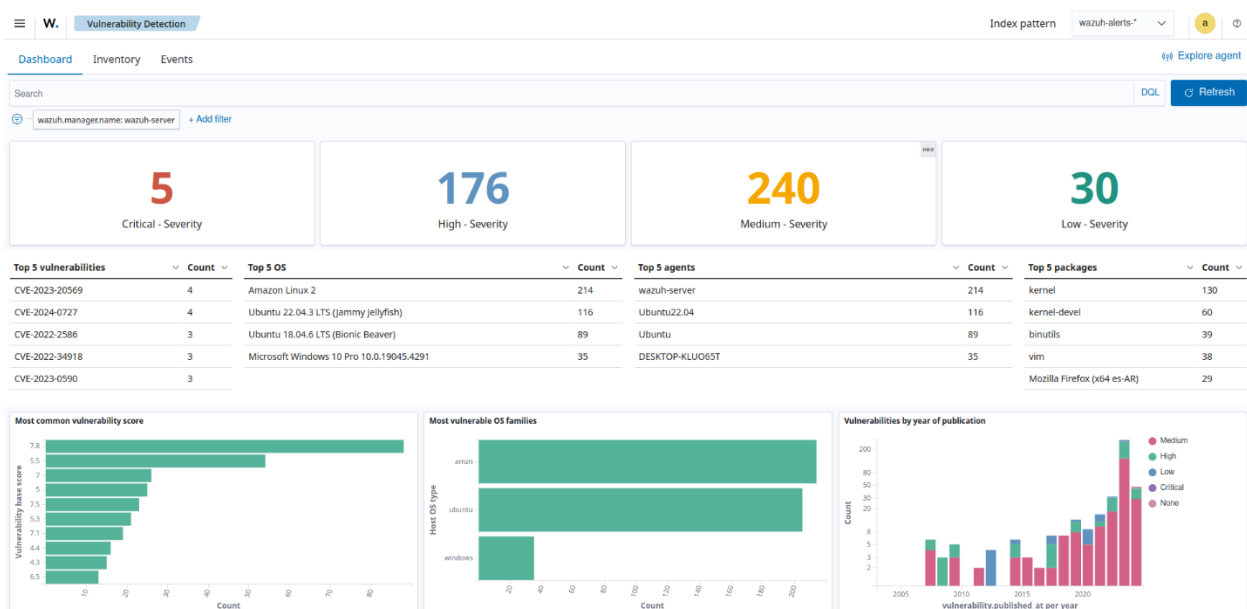
Subsequently, we merge this content, uploading the compiled documents to a cloud server. Finally, we publish these documents to our CTI API.

Relying on the Wazuh CTI, the [Vulnerability Detection](#) module supports a variety of operating systems, such as Windows, CentOS, Red Hat Enterprise Linux, Ubuntu, Debian, Amazon Linux, Arch Linux, and macOS operating systems, and applications.

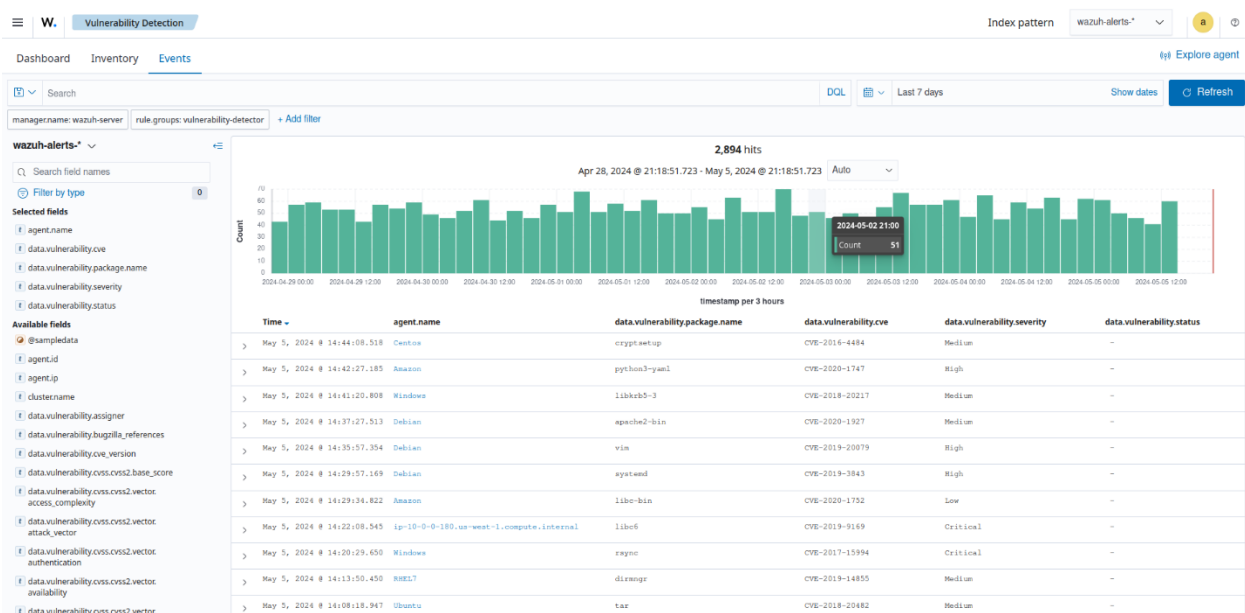
Achieve comprehensive visibility [Permalink to this headline](#)

The Vulnerability Detection module generates alerts for vulnerabilities discovered on the operating system and applications installed on the monitored endpoint. It correlates the software inventory collected by the Wazuh agent with the vulnerability content documents and displays the alert generated on the Wazuh dashboard. This provides a clear and comprehensive view of vulnerabilities identified in all monitored endpoints, allowing you to view, analyze and fix vulnerabilities.

The vulnerability detection dashboard shows the frequency of occurrences in different categories such as package name, operating system, agent name, vulnerability ID, and alert severity. This allows analysts to direct their focus appropriately.



You can view the alerts generated on the dashboard when new vulnerabilities are discovered.



The alerts generated on the dashboard could also be a result of remediation activities. The image below shows alerts generated after an upgrade or an uninstallation of a package resolved a vulnerability.



[Obtain actionable intelligence from vulnerability alerts](#)[Permalink to this headline](#)

Wazuh vulnerability alerts contain relevant information about the identified vulnerability which can help users understand and decide on remediation steps. You can see an example of a vulnerability detection alert below:

W.

Vulnerability Detection

Dashboard

Inventory

Events

Index pattern

wazuh-alerts-

Search

DQL

Last 7 days

Show dates

Refresh

manager.name: wazuh server

rule.groups: vulnerability-detector

Add filter

wazuh-alerts-

Search field names

Filter by type

0

Selected fields

data.vulnerability.cve

data.vulnerability.package.name

data.vulnerability.severity

data.vulnerability.status

Available fields

@sampledata

agent.id

agent.ip

agent.name

cluster.name

data.vulnerability.assigner

data.vulnerability.bugzilla_references

data.vulnerability.cve_version

data.vulnerability.cvs.cvs2.base_score

data.vulnerability.cvs.cvs2.vector.access_complexity

data.vulnerability.cvs.cvs2.vector.attack_vector

data.vulnerability.cvs.cvs2.vector.authentication

data.vulnerability.cvs.cvs2.vector.availability

data.vulnerability.cvs.cvs2.vector.confidentiality_impact

data.vulnerability.cvs.cvs2.vector.integrity_impact

data.vulnerability.cvs.cvs3.base_score

data.vulnerability.cvs.cvs3.vector.access_complexity

data.vulnerability.cvs.cvs3.vector.attack_vector

data.vulnerability.cvs.cvs3.vector.availability

data.vulnerability.cvs.cvs3.vector.confidentiality_impact

data.vulnerability.cvs.cvs3.vector.integrity_impact

data.vulnerability.cvs.cvs3.vector.privilege_required

data.vulnerability.cvs.cvs3.vector.scope

data.vulnerability.cve_reference

data.vulnerability.enumeration

data.vulnerability.package.architecture

data.vulnerability.package.condition

data.vulnerability.package.source

data.vulnerability.package.version

data.vulnerability.published

data.vulnerability.rationale

data.vulnerability.reference

data.vulnerability.references

data.vulnerability.state

data.vulnerability.title

data.vulnerability.type

data.vulnerability.updated

decoder.name

id

input.type

location

manager.name

rule.description

rule.fixedtimes

rule.gdpr

rule.groups

rule.id

rule.level

rule.mail

rule.pci_dss

rule.tsc

timestamp

2,892 hits

Apr 28, 2024 @ 21:26:01.008 - May 5, 2024 @ 21:26:01.008

Auto

Time

data.vulnerability.package.name

data.vulnerability.cve

data.vulnerability.severity

data.vulnerability.status

May 5, 2024 @ 14:44:08.518 cryptsetup CVE-2016-4484 Medium

Expanded document

View surrounding documents

View single document

Table

JSON

data.vulnerability.assigner

data.vulnerability.bugzilla_references

data.vulnerability.cve

data.vulnerability.cve_version

data.vulnerability.cvs.cvs2.base_score

data.vulnerability.cvs.cvs2.vector.access_complexity

data.vulnerability.cvs.cvs2.vector.attack_vector

data.vulnerability.cvs.cvs2.vector.authentication

data.vulnerability.cvs.cvs2.vector.availability

data.vulnerability.cvs.cvs2.vector.confidentiality_impact

data.vulnerability.cvs.cvs2.vector.integrity_impact

data.vulnerability.cvs.cvs3.base_score

data.vulnerability.cvs.cvs3.vector.access_complexity

data.vulnerability.cvs.cvs3.vector.attack_vector

data.vulnerability.cvs.cvs3.vector.availability

data.vulnerability.cvs.cvs3.vector.confidentiality_impact

data.vulnerability.cvs.cvs3.vector.integrity_impact

data.vulnerability.cvs.cvs3.vector.privilege_required

data.vulnerability.cvs.cvs3.vector.scope

data.vulnerability.cve_reference

data.vulnerability.enumeration

data.vulnerability.package.architecture

data.vulnerability.package.condition

data.vulnerability.package.source

data.vulnerability.package.version

data.vulnerability.published

data.vulnerability.rationale

data.vulnerability.reference

data.vulnerability.references

data.vulnerability.state

data.vulnerability.title

data.vulnerability.type

data.vulnerability.updated

decoder.name

id

input.type

location

manager.name

rule.description

rule.fixedtimes

rule.gdpr

rule.groups

rule.id

rule.level

rule.mail

rule.pci_dss

rule.tsc

timestamp

sampledata

._index

agent.id

agent.ip

agent.name

cluster.name

data.vulnerability.assigner

data.vulnerability.bugzilla_references

data.vulnerability.cve

data.vulnerability.cve_version

data.vulnerability.cvs.cvs2.base_score

data.vulnerability.cvs.cvs2.vector.access_complexity

data.vulnerability.cvs.cvs2.vector.attack_vector

data.vulnerability.cvs.cvs2.vector.authentication

data.vulnerability.cvs.cvs2.vector.availability

data.vulnerability.cvs.cvs2.vector.confidentiality_impact

data.vulnerability.cvs.cvs2.vector.integrity_impact

data.vulnerability.cvs.cvs3.base_score

data.vulnerability.cvs.cvs3.vector.access_complexity

data.vulnerability.cvs.cvs3.vector.attack_vector

data.vulnerability.cvs.cvs3.vector.availability

data.vulnerability.cvs.cvs3.vector.confidentiality_impact

data.vulnerability.cvs.cvs3.vector.integrity_impact

data.vulnerability.cvs.cvs3.vector.privilege_required

data.vulnerability.cvs.cvs3.vector.scope

data.vulnerability.cve_reference

data.vulnerability.enumeration

data.vulnerability.package.architecture

data.vulnerability.package.condition

data.vulnerability.package.source

data.vulnerability.package.version

data.vulnerability.published

data.vulnerability.rationale

data.vulnerability.reference

data.vulnerability.references

data.vulnerability.state

data.vulnerability.title

data.vulnerability.type

data.vulnerability.updated

decoder.name

id

input.type

location

manager.name

rule.description

rule.fixedtimes

rule.gdpr

rule.groups

rule.id

rule.level

rule.mail

rule.pci_dss

rule.tsc

timestamp

65

```

{
  "_index": "wazuh-alerts-4.x-sample-threat-detection",
  "_id": "e2ffSY8Be9PWdpLhA_nt",
  "_version": 1,
  "_score": null,
  "_source": {
    "predecoder": {},
    "cluster": {
      "name": "wazuh"
    },
    "agent": {
      "ip": "197.17.1.4",
      "name": "Centos",
      "id": "005"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "vulnerability": {
        "severity": "Medium",
        "package": {
          "condition": "Package less or equal than 2.1.7.3-2",
          "name": "cryptsetup",
          "version": "2:1.6.6-5ubuntu2.1",
          "architecture": "amd64"
        }
      },
      "references": [
        "http://hmarco.org/bugs/CVE-2016-4484/CVE-2016-4484_cryptsetup_initrd_shell.html",
        "http://www.openwall.com/lists/oss-security/2016/11/14/13",
        "http://www.openwall.com/lists/oss-security/2016/11/15/1",
        "http://www.openwall.com/lists/oss-security/2016/11/15/4",
        "http://www.openwall.com/lists/oss-security/2016/11/16/6",
        "http://www.securityfocus.com/bid/94315",

```

```

"https://gitlab.com/cryptsetup/cryptsetup/commit/ef8a7d82d8d3716ae9b58179590f7908981fa0c
b",
  "https://nvd.nist.gov/vuln/detail/CVE-2016-4484",
  "http://people.canonical.com/~ubuntu-security/cve/2016/CVE-2016-4484.html",
  "https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2016-4484"
],
"cve_version": "4.0",
"assigner": "cve@mitre.org",
"published": "2017-01-23",
"cwe_reference": "CWE-287",
"title": "CVE-2016-4484 on Ubuntu 16.04 LTS (xenial) - low.",
"rationale": "The Debian initrd script for the cryptsetup package 2:1.7.3-2 and earlier allows
physically proximate attackers to gain shell access via many log in attempts with an invalid
password.",
"cve": "CVE-2016-4484",
"state": "Fixed",
"bugzilla_references": [
  "https://launchpad.net/bugs/1660701"
],
"cvss": {
  "cvss2": {
    "base_score": "7.200000",
    "vector": {
      "integrity_impact": "complete",
      "confidentiality_impact": "complete",
      "availability": "complete",
      "attack_vector": "local",
      "access_complexity": "low",
      "authentication": "none"
    }
  },
  "cvss3": {
    "base_score": "6.800000",
    "vector": {
      "user_interaction": "none",

```

```

    "integrity_impact": "high",
    "scope": "unchanged",
    "confidentiality_impact": "high",
    "availability": "high",
    "attack_vector": "physical",
    "access_complexity": "low",
    "privileges_required": "none"
  }
},
  "updated": "2017-01-26"
},
"@sampledata": true,
"rule": {
  "firedtimes": 290,
  "mail": false,
  "level": 7,
  "pci_dss": [
    "11.2.1",
    "11.2.3"
  ],
  "tsc": [
    "CC7.1",
    "CC7.2"
  ],
  "description": "CVE-2016-4484 affects cryptsetup",
  "groups": [
    "vulnerability-detector"
  ],
  "id": "23504",
  "gdpr": [
    "IV_35.7.d"
  ]
},
"location": "vulnerability-detector",

```

```
"id": "1580123327.49031",
"decoder": {
  "name": "json"
},
"timestamp": "2024-05-05T17:44:08.518+0000"
},
"fields": {
  "data.vulnerability.published": [
    "2017-01-23T00:00:00.000Z"
  ],
  "data.vulnerability.updated": [
    "2017-01-26T00:00:00.000Z"
  ],
  "timestamp": [
    "2024-05-05T17:44:08.518Z"
  ]
},
"highlight": {
  "manager.name": [
    "@opensearch-dashboards-highlighted-field@wazuh-server@/opensearch-dashboards-highlighted-field@"
  ],
  "rule.groups": [
    "@opensearch-dashboards-highlighted-field@vulnerability-detector@/opensearch-dashboards-highlighted-field@"
  ]
},
"sort": [
  1714931048518
]
}
```

As you can see above, the alert contains key information about the detected vulnerability. This information includes the CVE information, reference links for further research, and a description that provides a concise explanation of the vulnerability.

Track vulnerability remediation[Permalink to this headline](#)

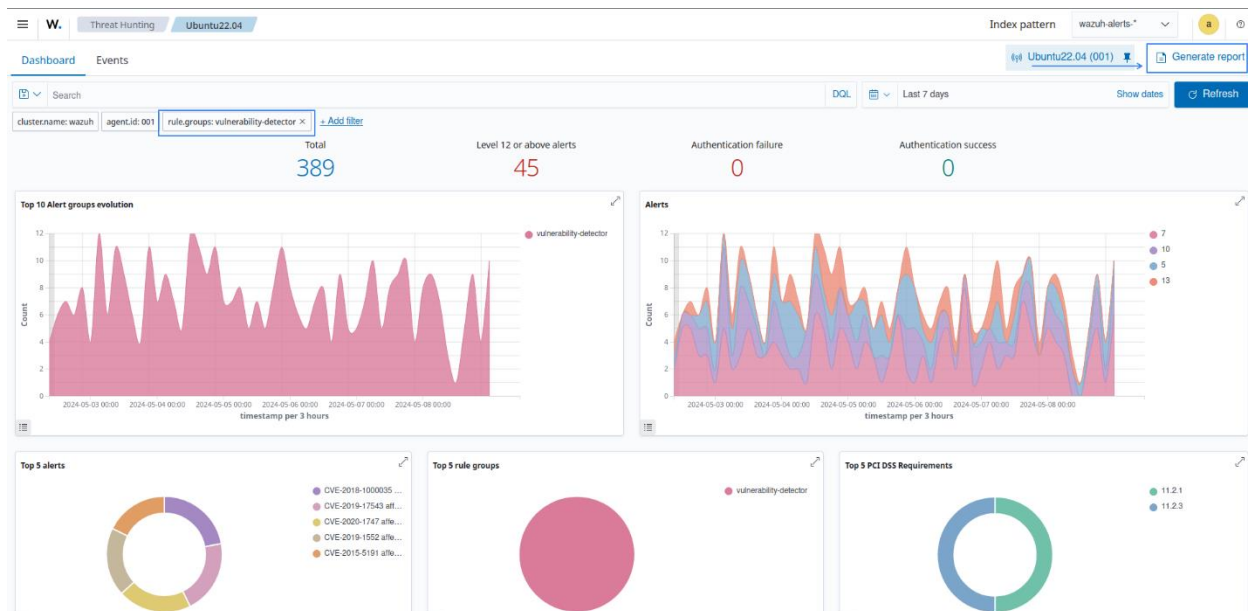
The Wazuh Vulnerability Detection module also allows you to confirm when a vulnerability has been remediated. This feature detects when a patch or software upgrade resolves a previously detected vulnerability. The feature is enabled using the [hotfixes](#) option and is available for Windows endpoints.

The screenshot displays the Wazuh Vulnerability Detection interface. On the left, a sidebar lists rule groups, rule IDs, rule levels, rule mail, rule pci_dss, rule tsc, and timestamp. The main panel shows a vulnerability alert for CVE-2021-24113, which is marked as 'Solved'. The alert details include the agent ID (001), agent IP (192.168.0.154), agent name (Windows11), and the vulnerability's CVSS base score (5.800000). The vulnerability is identified as a CVE affecting Microsoft Edge, with a package version of 124.0.2478.67. The alert was published on Feb 25, 2021, at 20:15:16.000. The reference URL is https://portal.msc.microsoft.com/en-US/security-guidance/advisory/CVE-2021-24113. The severity is Medium, and the status is solved. The title of the alert is 'CVE-2021-24113 affecting Microsoft Edge was solved'. The type is Packages, and it was updated on Dec 29, 2023, at 14:15:59.000.

Field	Value
data.vulnerability.cve	CVE-2021-24113
data.vulnerability.cvss.cvss2.base_score	5.800000
data.vulnerability.enumeration	CVE
data.vulnerability.package.architecture	x64
data.vulnerability.package.name	Microsoft Edge
data.vulnerability.package.version	124.0.2478.67
data.vulnerability.published	Feb 25, 2021 @ 20:15:16.000
data.vulnerability.reference	https://portal.msc.microsoft.com/en-US/security-guidance/advisory/CVE-2021-24113
data.vulnerability.severity	Medium
data.vulnerability.status	Solved
data.vulnerability.title	CVE-2021-24113 affecting Microsoft Edge was solved
data.vulnerability.type	Packages
data.vulnerability.updated	Dec 29, 2023 @ 14:15:59.000

Use vulnerability reports to identify critical security issues[Permalink to this headline](#)

Wazuh provides users with the ability to download a report that contains security events related to discovered and resolved vulnerabilities. This feature allows users to identify endpoints with unresolved vulnerabilities and keep track of remediation activities.



Incident response[Permalink to this headline](#)

A security incident refers to any adverse event or activity that risks or threatens the confidentiality, integrity, or availability of digital assets, networks, data, or resources. Such incidents include unauthorized access, data breaches, malware infections, denial-of-service attacks, and any other activities that compromise the security posture of an organization's information technology environment.

The goal of incident response is to effectively handle a security incident and restore normal business operations as quickly as possible. As organizations' digital assets continuously grow, managing incidents manually becomes increasingly challenging, hence the need for automation.

Automated incident response involves automatic actions taken when responding to security incidents. These actions can include isolating compromised endpoints, blocking malicious IP addresses, quarantining infected devices, or disabling compromised user accounts. By

1prevent or minimize the impact of incidents, and efficiently handle a large volume of security events.

Wazuh Active Response module[Permalink to this headline](#)

The Wazuh [Active Response](#) module allows users to run automated actions when incidents are detected on endpoints. This improves an organization's incident response processes, enabling security teams to take immediate and automated actions to counter detected threats.

You can also configure the actions to be either stateless or stateful. Stateless active responses are one-time actions while stateful responses revert their actions after some time.

Default active response actions[Permalink to this headline](#)

Out-of-the-box scripts are available on every operating system that runs the Wazuh agents. Some of the [default active response](#) scripts include:

Name of script	Description
disable-account	Disables a user account
firewall-drop	Adds an IP address to the iptables deny list.
firewalld-drop	Adds an IP address to the firewalld drop list.
restart.sh	Restarts the Wazuh agent or server.
netsh.exe	Blocks an IP address using netsh.

Custom active response actions[Permalink to this headline](#)

One of the benefits of the Wazuh Active Response module is its adaptability. Wazuh allows security teams to create **custom active response** actions in any programming language, tailoring them to their specific needs. This ensures that when a threat is detected, the response can be customized to align with the organization's requirements.

Automating incident response with Wazuh[Permalink to this headline](#)

To leverage the Wazuh Active Response module, you need to **configure** the action to be carried out when a specific event occurs on a monitored endpoint. For example, you can configure the Wazuh Active Response module to delete a malicious executable from an infected endpoint. In the examples that follow, we show how the Wazuh Active Response module handles different incidents.

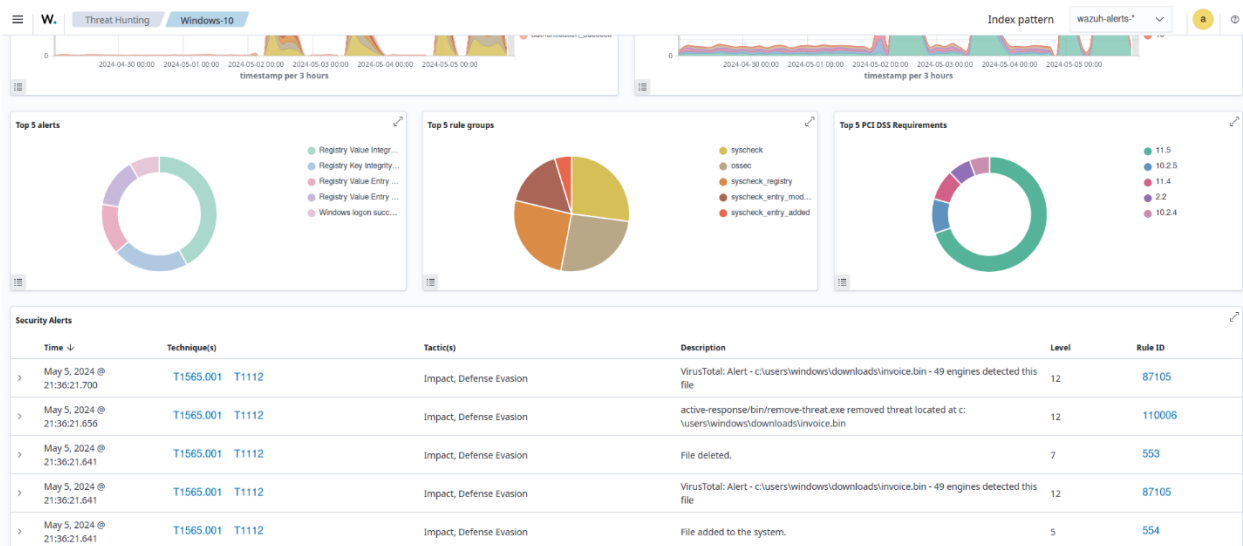
Removing malware[Permalink to this headline](#)

You can use the Wazuh Active Response module in conjunction with the **File Integrity Monitoring** module and **VirusTotal integration** to detect and remove malicious files from an endpoint.

The image below shows the following activities:

1. Rule ID **554** is fired when a file is added to the **Downloads** directory which is monitored with the Wazuh File Integrity Monitoring module.
2. Rule ID **87105** triggers when Wazuh extracts the file hash, requests data about the file hash from the VirusTotal database via its API, and receives a malicious file response.

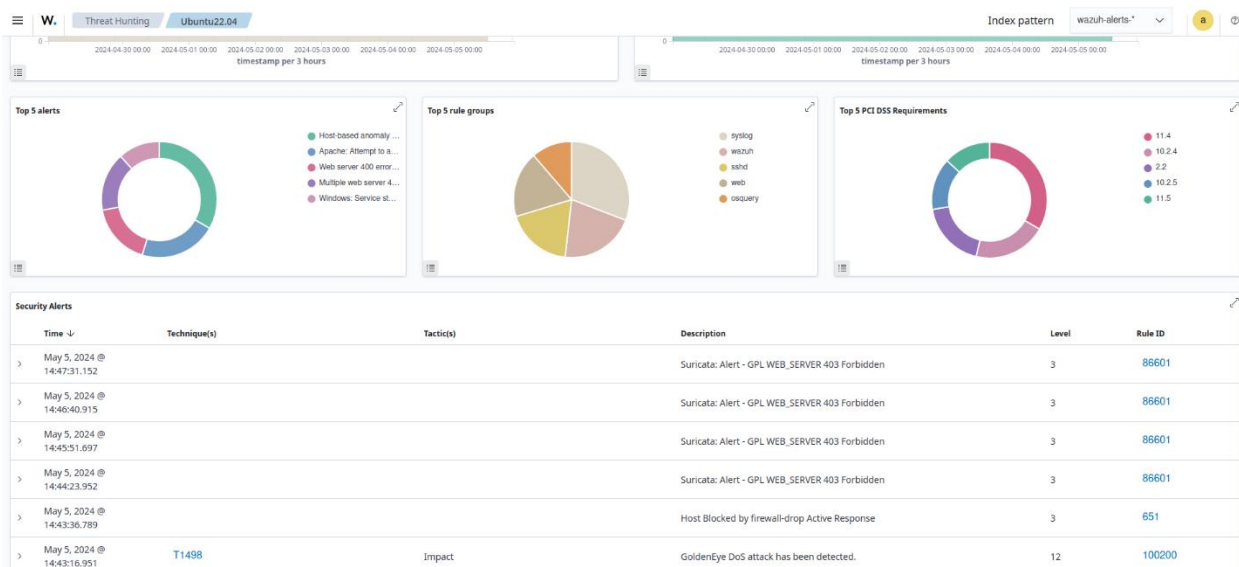
3. Rule ID **553** is fired when a file is deleted from the **Downloads** directory which is monitored with the Wazuh File Integrity Monitoring module.
4. Rule ID **110006** is fired when the Wazuh Active Response module deletes the malicious file from the endpoint.



In this scenario, the Wazuh Active Response module automatically removes the malicious file, reducing the time between threat detection and mitigation.

Responding to DoS attacks [Permalink to this headline](#)

The primary goal of a DoS attack is to render the target inaccessible to legitimate users, causing a denial of service. In the image below, we show how the Wazuh Active Response module blocks malicious IP addresses performing a DoS against a web server on an Ubuntu endpoint.

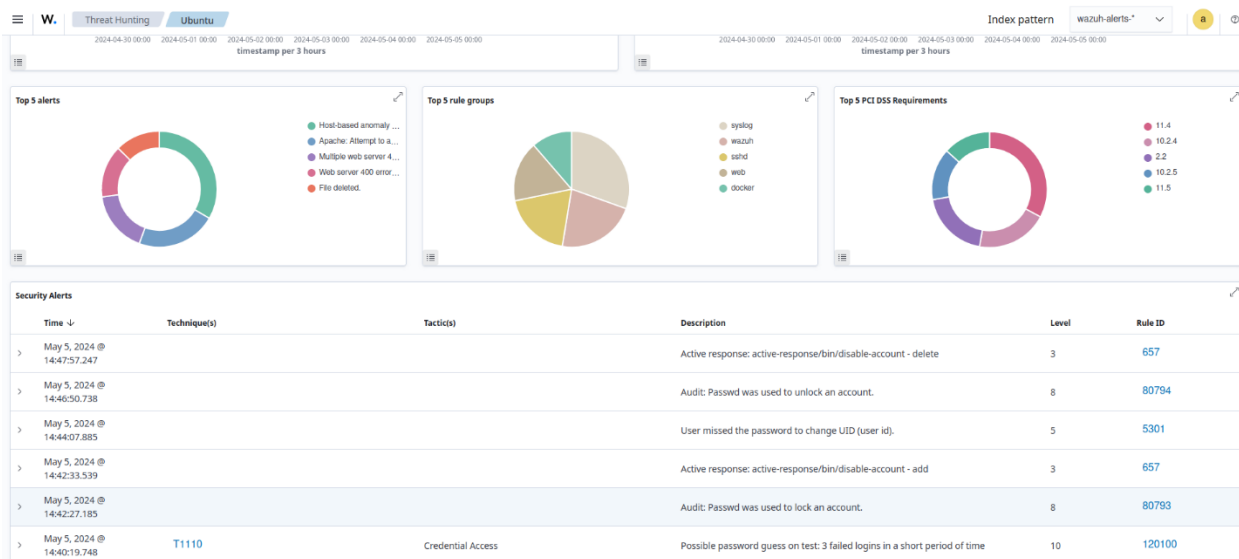


In this case, the Wazuh Active Response module automatically blocks the malicious hosts from causing a DoS attack on the web server. Thereby ensuring the availability of the web server to the authorized users.

Disabling a user account after a brute-force attack [Permalink to this headline](#)

Account lockout is a security measure used to defend against brute force attacks by limiting the number of login attempts a user can make within a specified time. We use the Wazuh Active Response module to disable the user account whose password is being guessed by an attacker.

In the image below, the Wazuh Active Response module disables the account on a Linux endpoint and re-enables it again after 5 minutes.



In this scenario, when an attacker tries to guess a user's password repeatedly and fails, the account becomes temporarily inaccessible. This impedes attackers who rely on brute-force methods to guess user account passwords.

By utilizing the Wazuh Active Response module, security teams can automate responses to different incidents. Thereby ensuring efficient incident response and a more resilient cybersecurity posture.